

1. INTRODUCTION TO AVIATION

1.1 History of Aviation

Aviation is the design, development, production, operation, and use of aircraft, especially heavier-than-air aircraft. Aviation is derived from avis, the Latin word for bird. The history of aviation has extended over more than two thousand years from the earliest attempts in kites and gliders to powered heavier than air supersonic and hypersonic flight. Aviation term applied to the science and practice of flight in heavier than air craft, including airplanes, gliders, helicopters, addend VTOL (vertical takeoff and landing) and STOL (short takeoff and landing) craft. These are distinguished from lighter than air craft, which include balloons, airships etc.

Operational aviation is grouped broadly into three classes: military aviation, commercial aviation and general aviation. Military aviation is the use of aircraft and other flying machines for the purposes of conducting or enabling warfare, including national airlift (cargo) capacity to provide logistical supply to forces stationed in a theater or along a front. Air power includes the national means of conducting such warfare including the intersection of transport and war craft. The wide variety of military aircraft includes bombers, fighters, and fighter bombers. Commercial aviation includes primarily the operation of scheduled and charter airlines. General aviation may include business flights, private aviation, flight training, ballooning, parachuting, gliding, hang gliding, aerial photography, foot-launched powered hang gliders, air ambulance, crop dusting, charter flights, traffic reporting, police air patrols and forest fire fighting..

1.2 Aircraft

An aircraft is a machine that is able to fly by gaining support from the air, or, in general, the atmosphere of a planet. It counters the force of gravity by using either static lift or by using the dynamic lift of an airfoil or in a few cases the downward thrust from jet engines. In another word aircraft is weight carrying structure that can travel through the air, supported either by its own buoyancy or by the dynamic action of air against its surfaces. The human activity that surrounds aircraft is called aviation. Crewed aircraft are flown by an onboard pilot, but unmanned aerial vehicles may be remotely

controlled or self-controlled by onboard computers. Aircraft may be classified by different criteria, such as lift type, propulsion, usage, and others.

1.3 Airport

An airport is a location where aircraft such as fixed-wing aircraft, helicopters, and blimps take off and land. Aircraft may be stored or maintained at an airport. An airport consists of at least one surface such as a runway for a plane to take off and land, a helipad, or water for takeoffs and landings, and often includes buildings such as control towers, hangars and terminal buildings. Larger airports may have fixed base operator services, seaplane docks and ramps, air traffic control, passenger facilities such as restaurants and lounges, and emergency services. A military airport is known as an airbase or air station. A water airport is a water aerodrome (an area of open water used regularly by seaplanes or amphibious aircraft for landing and taking off), usually with passenger facilities on adjacent land, which acts as an airport.

1.4 Airspace

Airspace means the portion of the atmosphere controlled by a country above its territory, including its territorial waters or, more generally, any specific three-dimensional portion of the atmosphere. Airspace may be further subdivided into a variety of areas and zones, including those where there are either restrictions on flying activities or complete prohibition of flying activities. By international law, the notion of a country's sovereign airspace corresponds with the maritime definition of territorial waters as being 12 nautical miles (22.2 km) out from a nation's coastline.

1.5 Air traffic Controller

Air traffic control (ATC) involves communication with aircraft to help maintain separation that is they ensure that aircraft are sufficiently far enough apart horizontally or vertically for no risk of collision. Controllers may co-ordinate position

reports provided by pilots, or in high traffic areas (such as the United States) they may use radar to see aircraft positions.

ATC is especially important for aircraft flying under Instrument flight rules (IFR), where they may be in weather conditions that do not allow the pilots to see other aircraft. However, in very high-traffic areas, especially near major airports, aircraft flying under Visual flight rules (VFR) are also required to follow instructions from ATC.

There are generally four different types of ATC:

- Center controllers, who control aircraft en route between airports
- Control towers (including tower, ground control, clearance delivery, and other services), which control aircraft within a small distance (typically 10–15 km horizontal, and 1,000 m vertical) of an airport.
- Oceanic controllers, who control aircraft over international waters between continents, generally without radar service.
- Terminal controllers, who control aircraft in a wider area (typically 50–80 km) around busy airports.

1.5 Air Traffic Management

ATM is the regulation of air traffic in order to avoid exceeding airport or air traffic control capacity in handling traffic, and to ensure that available capacity is used efficiently. Because only one aircraft can land or depart from a runway at the same time, and because aircraft must be separated by certain time to avoid collisions, every airport has a finite capacity; it can only safely handle so many aircraft per hour. This capacity depends on many factors, such as the number of runways available, layout of taxi tracks, availability of air traffic control, and current or anticipated weather. Especially the weather can cause large variations in capacity because strong winds may limit the number of runways available, and poor visibility may necessitate increases in separation between aircraft. Air traffic control can also be limiting, there are only so many aircraft an air traffic control unit can safely handle. Staff shortages, radar maintenance or equipment faults can lower the capacity of a unit. This can affect both airport air traffic control as well as en-route air traffic control center.

2. AERONAUTICAL COMMUNICATION

2.1. Aviation Band

Aviation Band or Airband or Aircraft band is the name for a group of frequencies in the VHF radio spectrum allocated to radio communication in civil aviation, sometimes also referred to as *VHF*, or phonetically as "*Victor*". Different sections of the band are used for radionavigational aids and air traffic control.

The VHF airband uses the frequencies between 108 and 137 MHz. The lowest 10 MHz of the band, from 108–117.95 MHz, is split into 200 narrow-band channels of 50 kHz. These are reserved for navigational aids such as VOR beacons, and precision approach systems such as ILS localizers.

As of 2012, most countries divide the upper 19 MHz into 760 channels for amplitude modulation voice transmissions, on frequencies from 118–136.975 MHz, in steps of 25 kHz. In Europe, it is becoming common to further divide those channels into three (8.33 kHz channel spacing), potentially permitting 2,280 channels. Some channels between 123.100 and 135.950 are available in the US to other users such as government agencies, commercial company advisory, search and rescue, military aircraft, glider and ballooning air-to-ground, flight test and national aviation authority use. A typical transmission range of an aircraft flying at cruise altitude (35,000 ft), is about 200 miles in good weather conditions.

Aeronautical voice communication is also conducted in other frequency bands, including satellite voice on Inmarsat and high frequency voice in the North Atlantic and remote areas. Military aircraft also use a dedicated UHF-AM band from 220.0–399.95 MHz for air-to-air and air-to-ground, including air traffic control communication. This band has a designated emergency and guard channel of 243.0 MHz.

Some types of nav aids, such as Non-directional beacons and Distance Measuring Equipment, do not operate on these frequencies; in the case of NDBs the Low frequency and Medium frequency bands are used between 190–415 kHz and 510–535 kHz. The ILS glide path operates in the UHF frequency range of 329.3–335.0 MHz, and DME also uses UHF from 962–1150 MHz.

2.2. ICAO and ITU

2.2.1. ICAO

The International Civil Aviation Organization (ICAO) is a specialized agency of the United Nations. It codifies the principles and techniques of international air navigation and fosters the planning and development of international air transport to ensure safe and orderly growth. Its headquarters are located in the *Quartier International* of Montreal, Quebec, Canada.

The ICAO Council adopts standards and recommended practices concerning air navigation, its infrastructure, flight inspection, prevention of unlawful interference, and facilitation of border-crossing procedures for international civil aviation. In addition, the ICAO defines the protocols for air accident investigation followed by transport safety authorities in countries signatory to the Convention on International Civil Aviation, commonly known as the *Chicago Convention*.

The Air Navigation Commission (ANC) is the technical body within ICAO. The Commission is composed of 19 Commissioners, appointed by the Council. Commissioners serve as independent experts, who although nominated by their states, do not serve as state or political representatives. The development of Aviation Standards and Recommended Practices is done under the direction of the ANC through the formal process of ICAO Panels. Once approved by the Commission, standards are sent to the Council, the political body of ICAO, for consultation and coordination with the Member States before final adoption.

The ICAO should not be confused with the International Air Transport Association (IATA), a trade organization for airlines also headquartered in Montreal, or with the Civil Air Navigation Services Organisation (CANSO), an organization for Air Navigation Service Providers (ANSPs) with its headquarters at Amsterdam Airport Schiphol in the Netherlands. These are trade associations representing specific aviation interests, whereas ICAO is a body of the United Nations.

2.2.2. ITU

The International Telecommunication Union, originally founded as the International Telegraph Union, is a specialized agency of the United Nations concerned with

Information and Communication Technologies (ICTs). As such, it is responsible for the allocation of radio spectrum and satellite orbits, and for the standardization and development of ICTs worldwide. ITU also organizes worldwide and regional exhibitions and forums, such as ITU TELECOM WORLD, bringing together representatives of government and the telecommunications and ICT industry to exchange ideas, knowledge and technology.

The ITU, based in Geneva, Switzerland, is a member of the United Nations Development Group. Founded on the principle of international cooperation between governments (Member States) and the private sector (Sector Members, Associates and Academia), ITU is the premier global forum through which parties work towards consensus on a wide range of issues affecting the future direction of the ICT industry.

The ITU comprises three sectors, each managing a different aspect of the matter handled by the Union, as well as ITU Telecom:

➤ **Radio communication (ITU-R)**

- Managing the international radio-frequency spectrum and satellite orbit resources is at the heart of the work of the ITU Radio communication Sector (ITU-R).

➤ **Standardization (ITU-T)**

- ITU's standards-making efforts are its best-known and oldest activity; known prior to 1992 as the International Telephone and Telegraph Consultative Committee or CCITT

➤ **Development (ITU-D)**

- Established to help spread equitable, sustainable and affordable access to information and communication technologies (ICT).

ITU Membership

Membership of ITU is open to governments, which may join the Union as Member States, as well as to private organizations like carriers, equipment manufacturers,

funding bodies, research and development organizations and international and regional telecommunication organizations, which can join ITU as non-voting Sector Members

2.3 VHF air to ground communication

The modernization of air/ground (A/G) communications in the very high frequency (VHF) band for Air Traffic Services (ATSS) and Aeronautical Operational Control (AOC) applications is becoming increasingly important as the competition for the existing frequency spectrum intensifies. The Aviation Industry currently exploring various modernization alternatives ranging from channel splitting to an integrated digital voice and data system utilizing a single avionics unit that supports ATS and AOC. A mandatory aspect of ATS applications is the preservation of real-time response capabilities for critical services.

Aircraft Communications Addressing and Reporting System (ACARS) is a digital datalink system for transmission of short, relatively simple messages between aircraft and ground stations via radio or satellite. The protocol, which was designed by Aeronautical Radio, Incorporated (ARINC) to replace their very high frequency (VHF) voice service and deployed in 1978, uses telex formats. SITA later augmented their worldwide ground data network by adding radio stations to provide ACARS service. Over the next 20 years, ACARS will be superseded by the Aeronautical Telecommunications Network (ATN) protocol for Air Traffic Control communications and by the Internet Protocol for airline communications.

A person or a system on board may create a message and send it via ACARS to a system or user on the ground, and vice versa. Messages may be sent either automatically or manually.

VHF subnet work

A network of VHF ground radio stations ensure that aircraft can communicate with ground end systems in real-time from practically anywhere in the world. VHF communication is line-of-sight, and provides communication with ground-based transceivers (often referred to as Remote Ground Stations or RGSs). The typical range is dependent on altitude, with a 200-mile transmission range common at high altitudes.

Thus VHF communication is only applicable over landmasses which have a VHF ground network installed.

SATCOM and HF sub networks

SATCOM can provide worldwide coverage. Depending on the satellite system in use, coverage may be limited or absent at high latitudes (such as needed for flights over the poles). HF datalink is a relatively new network whose installation began in 1995 and was completed in 2001. Aircraft with HF or global SATCOM data link can fly polar routes and maintain communication with ground based systems (ATC centers and airline operation centers). ARINC is the only service provider for HF data link.

2.4 HF ground to ground communication

High frequency (HF) radio provides aircraft with an effective means of communication over long distance oceanic and trans-polar routes. In addition, global data communication has recently been made possible using strategically located HF data link (HFDL) ground stations. These provide access to ARINC and SITA airline networks. HF communication is thus no longer restricted to voice and is undergoing a resurgence of interest due to the need to find a means of long distance data communication that will augment existing VHF and SATCOM data links. An aircraft HF radio system operates on spot frequencies within the HF spectrum. Unlike aircraft VHF radio, the spectrum is not divided into a large number of contiguous channels but aircraft allocations are interspersed with many other services, including short wave broadcasting, fixed point-to-point, marine and land-mobile, government and amateur services.

In the HF range (3 MHz to 30 MHz) radio waves propagate over long distances due to reflection from the ionised layers in the upper atmosphere. Due to variations in height and intensities of the ionised regions, different frequencies must be used at different times of day and night and for different paths. There is also some seasonal variation (particularly between winter and summer). Propagation may also be disturbed and enhanced during periods of intense solar activity. The upshot of this is that HF propagation has considerable vagaries and is far less predictable than propagation at VHF. Frequencies chosen for a particular radio path are usually set roughly mid-way between the Figure 5.1 VHF aircraft coverage in the North Atlantic area lowest usable frequency (LUF) and the maximum usable frequency (MUF).

The daytime LUF is usually between 4 to 6 MHz during the day, falling rapidly after sunset to around 2 MHz. The MUF is dependent on the season and sunspot cycle but is often between 8 MHz and 20 MHz. Hence a typical daytime frequency for aircraft communication might be 8 MHz whilst this might be as low as 3 MHz during the night. Typical ranges are in the region of 500 km to 2500 km and this effectively fills in the gap in VHF coverage.

2.5. AFTN/ATN/AMSH

2.5.1 AFTN

The **Aeronautical Fixed Telecommunication Network (AFTN)** is a worldwide system of aeronautical fixed circuits provided, as part of the Aeronautical Fixed Service, for the exchange of messages and/or digital data between aeronautical fixed stations having the same or compatible communications characteristics. AFTN comprises aviation entities including: ANS (Air Navigation Services) providers, aviation service providers, airport authorities and government agencies, to name a few. It exchanges vital information for aircraft operations such as distress messages, urgency messages, flight safety messages, meteorological messages, flight regularity messages and aeronautical administrative messages.

2.5.1.1 AFTN Station Format

An AFTN address is an eight-letter-group composed of a four-letter ICAO Location Indicator plus a three-letter-group identifying an organization or service addressed and an additional letter. The additional letter represents a department, division or process within the organization/function addressed. The letter *X* is used to complete the address when an explicit identification of the department, division or process is not required.

Examples for some common three-letter-groups used in AFTN addresses in order to identify an organization or service:

Facility code	refers to
YFYX	"AFTN Office"
ZTZX	"Control tower"
ZPZX	"ATS Reporting Office"
ZQZX	"Area Control Center"
YNYX	"Notam Office"
YDYX	"Airport Manager"
YZYX	"Met Data Bank"
YMYX	"Local Met Office"
ZRZA	"Radar Approach"

2.5.1.1 AFTN Message categories and priority indicator

➤ Messages

- Distress messages;
- Urgency messages;
- Flight safety messages;
- Meteorological messages;
- Flight regularity messages;
- Aeronautical information services (AIS) messages;
- Aeronautical administrative messages;
- Service messages.

➤ Priority indicator

- Priority Indicator SS for Distress Messages
- Priority Indicator DD for Urgency Messages
- Priority Indicator FF for Flight Safety Messages
- Priority Indicator GG for Meteorological Messages, Flight Regularity Messages and Aeronautical Information Services Messages
- Priority Indicator KK for Aeronautical Administrative Messages

- Priority Indicator used for Service Messages are assigned as considered appropriate by the originator, but most likely *KK* is used

2.5.3 AMHS

Aeronautical Message Handling System (AMHS) also is a standard for aeronautical ground-ground communications based on X.400 profiles. It has been defined by the ICAO. ICAO defines two fundamental levels of service within the ATSMHS;

- Basic ATSMHS and
- The Extended ATSMHS.

Additionally, ICAO outlines different subsets of the Extended ATSMHS. The Basic ATSMHS performs an operational role similar to the Aeronautical Fixed Telecommunication Network with a few enhancements. The Extended ATSMHS provided enhanced features but includes the Basic level of service capability; in this way it is ensured that users with Extended Service capabilities can interoperate, at a basic level, with users having Basic Service capabilities and vice-versa.

The ATSMHS is provided by a set of end systems, which collectively comprise the ATS Message Handling System. The systems co-operate to provide users (human or automated) with a data communication service. The AMHS network is composed of interconnected ATS Message Servers that perform message switching at the application layer.

3. AERONAUTICAL NAVIGATION

3.1 INDRODUCTION

3.1.1. Introduction to Navigation System

Finding the way from one place to another is called NAVIGATION. Moving of an aircraft from one point to another is the most important part for any kind of mission. Plotting on the paper or on the map a course towards a specific area of the earth, in the past, used to be a task assigned to a specialized member of the aircraft's crew such a navigator. Such a task was quite complicated and not always accurate. Since it depended on the observation, using simple maps and geometrical instruments for calculations. Today, aerial navigation has become an art which nears to perfection. Both external Nav aids (Navigational Aids) and on-board systems help navigate any aircraft over thousands of miles with such accuracy that could only be imagined a few decades ago.

3.1.2. Piloting

Piloting is the most common method of air navigation. This method, the pilot keeps on course by following a series of landmarks on the ground. Usually before take-off, pilot will making pre-flight planning, the pilot will draw a line on the aeronautical map to indicate the desired course. Pilot will notes various landmarks, such as highways, railroad tracks, rivers, bridges. As the pilot flies over each of landmark, pilot will checks it off on the chart or map. If the plane does not pass directly over the landmark, the pilot will know that he has to correct the course.

3.1.3. Dead Reckoning

Dead reckoning is the primary navigation method used in the early days of flying. It is the method on which Lindberg relied on his first trans-Atlantic flight. A pilot used this method when flying over large bodies of water, forest, deserts. It demands more skill and experience than pilot age does. It is based on time, distance, and direction only. The pilot must know the distance from one point to the next, the magnetic heading to be flown. Pilot works on the pre-flight plan chart, pilot plan a route in advance. Pilot calculates the time to know exactly to reach the destination while flying at constant speed. In the air,

the pilot uses compass to keep the plane heading in the right direction. Dead reckoning is not always a successful method of navigation because of changing wind direction. It is the fundamental of VFR flight

3.1.4. Radio Navigation

Radio navigation is used by almost all pilots. Pilots can find out from an aeronautical chart what radio station they should tune to in a particular area. They can then tune their radio navigation equipment to a signal from this station. A needle on the navigation equipment tells the pilot where they are flying to or from station, on course or not.

and the standby with 400Hz.

3.3 VHF Omi Directional Radio Range (VOR)

3.3.1 Navigational Aid

A navigation system in which one or more signals are emitted from a facility (or co-located facilities) to produce simultaneous indication of bearing and distance. Since a bearing is a radial line of position and a distance is a circular line of position, the rho-theta system always ensures a position fix produced by the intersection of two lines of position which are at right angles to each other. This produces a minimum geometric dilution of position, a figure of merit for all radio navigation systems, and is one of the chief advantages of a rho-theta system. Another major advantage is that it is a single-site system and can thus be installed on a ship or an island. This has also made it attractive politically, enabling small countries to have their own navigation systems. VOR is an example of such a system.

VOR as a Navigation Aid:

Accuracy: radial alignment error < 3 degrees - maximum amplitude of bend

Ground: monitors are placed around the site to detect drift in the radiated signal if signal exceeds tolerance, the transmitter shuts down.

Air: If signal level, and/or, if either of the modulation levels falls below a preset level an error flag signal is sent to the HSI

Availability: Most sites have dual transmitters so if one fails the other takes over. Availability is better than 99.

VOR

VOR, short for VHF omnidirectional radio range, is a type of short-range radio navigation system for aircraft, enabling aircraft to determine their position and stay on course by receiving radio signals transmitted by a network of fixed ground radio beacons, with a receiver unit. It uses radio frequencies in the very high frequency (VHF) band from 108 to 117.95 MHz. Developed in the US beginning in 1937 and deployed by 1946, VOR is the standard air navigational system in the world, used by both commercial and general aviation. There are about 3000 VOR stations around the world.

A VOR ground station sends out a master signal which is sent with equal intensity in all directions, and a highly directional second signal that varies in phase 30 times a second compared to the master. This signal is timed so that the phase varies as the secondary antenna spins, such that when the antenna is 90 degrees from north, the signal is 90 degrees out of phase of the master. By comparing the phase of the secondary signal to the master, the angle (bearing) to the station can be determined. This bearing is then displayed in the cockpit of the aircraft, and can be used to take a fix as in earlier radio direction finding (RDF) systems, although it is, in theory, easier to use and more accurate. This line of position is called the "radial" from the VOR. The intersection of two radials from different VOR stations on a chart provides the position of the aircraft. VOR stations are fairly short range, the signals have a range of about 200 miles.

VOR stations broadcast a VHF radio composite signal including the station's identifier, voice (if equipped), and navigation signal. The identifier is typically a two- or three-letter string in Morse code. The voice signal, if used, is usually the station name, in-flight recorded advisories, or live flight service broadcasts. The navigation signal allows the airborne receiving equipment to determine a magnetic bearing from the station to the aircraft (direction from the VOR station in relation to the Earth's magnetic North at the time of installation). VOR stations in areas of magnetic compass unreliability are oriented with respect to True North.

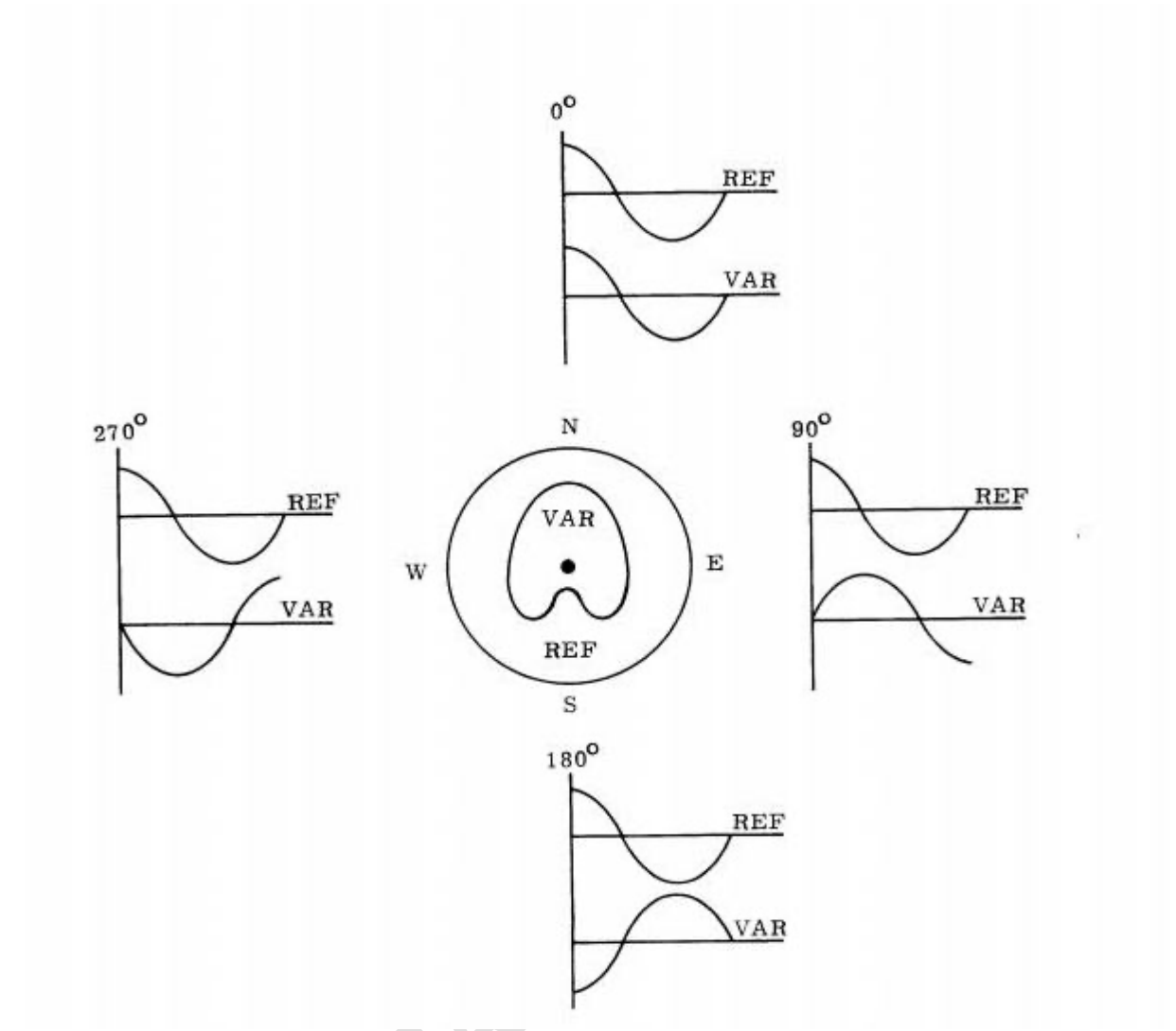
3.3.2. Frequency Band

- Operating frequency: 108MHz to 118 MHz
- Mainly used in frequency: 112MHz to 118MHz.
- Frequency used by Kathmandu VOR: 113.2 MHz

The channel separation between the VOR's is generally taken either 100 KHz or 200KHz. from place to place with frequency tolerance of $\pm 0.005\%$. In order to accommodate more VOR beacons, in some densely installed areas, the channel separation may be used as close as 50 KHz. But to avoid harmful interference or inter-modulation the frequency tolerance of all the nearby VOR's should be maintained at $\pm 0.002^\circ$.

3.3.3. General Principal of Operation

The principle of operation is bearing measurement by phase comparison. This means that the transmitter on the ground produces and transmits a signal, or actually two separate signals, which make it possible for the receiver to determine its position in relation to the ground station by comparing the phases of these two signals. In theory, the VOR produces a number of tracks all originating at the transmitter. These tracks are called radials and are numbered from 1 to 360, expressed in degrees, or $^\circ$. The 360° radial is the track leaving the VOR station towards the Magnetic North, and if you continue with the cardinal points, radial 090° points to the East, the 180° radial to the South and the 270° radial to the West, all in relation to the magnetic North. Think of a lighthouse at sea and imagine the white light rotating at a speed of one revolution per minute (60 seconds). Every time this white narrow beam passes through Magnetic North, a green omnidirectional light flashes. Omnidirectional means that it can be seen from any position around the lighthouse. If we are situated somewhere in the vicinity of the light sources and are able to see them, we can measure the time interval from the green light flash until we see the white light. The elapsed time is directly proportional to our position line in relation to the lighthouse. The speed of 1 RPM corresponds to 6° per second, so if 30 seconds elapse between the time we see the green flash and the white rotating light, we are on the 180° radial, or directly south of the station ($30 \text{ sec} \times 6^\circ/\text{sec} = 180^\circ$). This calculation can be done from any position and the elapsed time is directly proportional to our angular position (radial). We could name these light signals, calling the green one the Reference (REF) signal and the white beam the Variable (VAR) signal. The same principle is employed in a VOR.



3.3.4 VOR Transmission Techniques

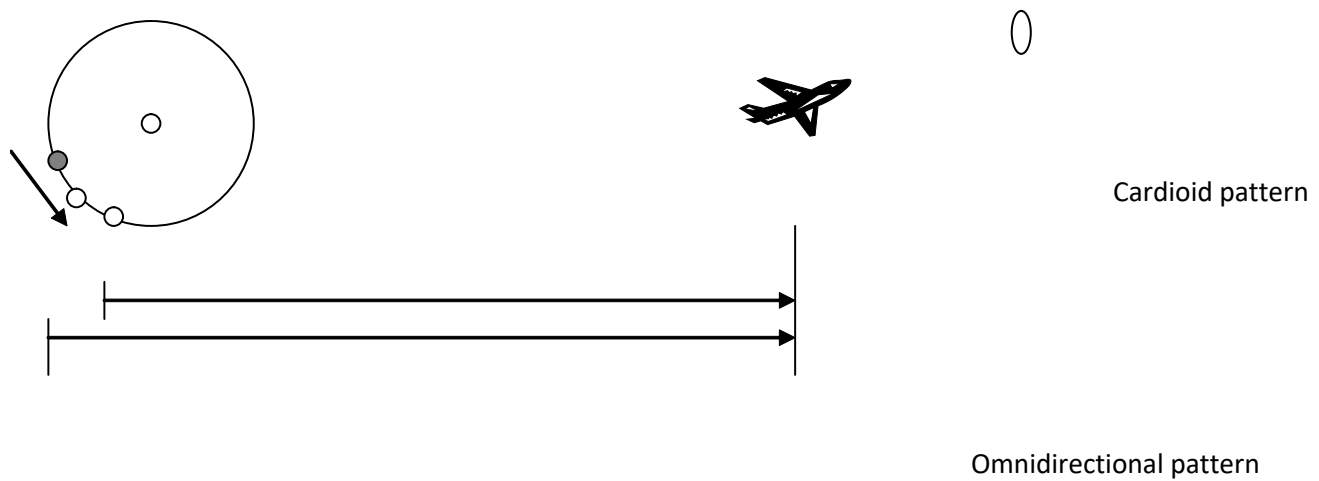


Figure: General block diagram of CVOR transmitter

9960 Hz generator produces the sub-carrier frequency, which is the basis for the REF signal. The 9960 Hz is fed to a 30Hz frequency modulator in order to produce the frequency deviation of ± 480 Hz. This again is amplitude modulated with 30% on the VHF transmitter, normally 50 to 200 Watts output. The REF signal is fed to an omnidirectional antenna, which normally is a loop antenna, called *Alford loop* antenna. The REF signal is also in parallel fed to a modulation eliminator, which removes the modulation, and the signal output from this block is then a clean continuous wave (CW) signal with the same carrier frequency. This signal is fed to the horizontally polarized dipole antenna to obtain the figure of eight pattern. Since the frequency of these two signals are the same, they will combine together to form a cardioid. The goniometer rotates the figure of eight at 1800 rpm, which will also cause the cardioid to rotate at the same rate.

A cardioid has maximum and minimum radiation pattern. While rotating, when the maximum pattern is towards the receiver it will receive maximum signal and for minimum pattern the signal received will be minimum. Therefore, if the cardioid is made to rotate at 1800 times per minute (30 times per second), the receiver will get the signal as 30 Hz AM. The following Fig. 2.2.6 explains the rotation of the cardioid pattern and the resulting AM signals received by the airborne VOR receivers at north, east, south and west directions. From these figures it is evident that the variable phase of the amplitude modulation (space modulation) is dependent of azimuth degree by degree.

VOR is a phase comparison system. This means simply that the phase of one signal is compared with the phase of another. However, a problem arises in that this type of comparison is possible only between signals whose frequencies are identical, and one also needs to be able to identify the source of each signal. Some-times this identification can be accomplished by time multiplexing the signals and storing the phase information in the receiver circuitry for later comparison. But in the VOR system both signals are transmitted simultaneously, and there needs to be a way to prevent the two from producing a resultant as they pass through space. This is accomplished by transmitting one signal as amplitude modulation and the other as frequency modulation. Detection in the

receiver produces two separate audio signals of exactly the same frequency but with measurable phase difference.

The two signals being compared are both 30 Hz signals. One is transmitted in a manner so as to produce a circular radiation pattern. Thus all aircraft at the same instant in time, and at the same distance from the transmitter, receive the same exact phase of this 30 Hz signal. Consequently, this signal is called the reference signal. The other signal has a radiation pattern shaped like a cardioid, and this pattern is caused to rotate about the station 30 times per second. The airborne equipment which receives this cardioid signal detects a signal strength which depends on which part of the cardioid it is receiving at any particular instant of time. By the rotation of this pattern amplitude modulation is created, and the received signal strength from the variable signal undergoes a cyclic change which is repeated 30 times per second.

Since both signals occur at the same 30 Hz rate there is a repetitive synchronization between the two, and all that is needed is an index point so that they may be compared for phase difference. This index point is established on the reference signal, and the phase of the variable signal is initially adjusted so that the phase difference between the two is 0° at magnetic north. The rotation of the variable signal from north azimuth creates a phase difference of one electrical degree for each rotational degree; therefore, an observer can determine his geographical azimuth by simply measuring the phase difference between the phase of the reference signal and the phase of the variable signal. A receiver anywhere within the coverage area of a facility will receive two 30 Hz signals—one from the reference field and one from the variable field cardioid that is passing by. The phase of the signal from the variable field will lag that of the reference field by the exact number of degrees the receiver bears from magnetic north.

3.3.6. VOR Errors

1) **Multi-path errors:** The major bearing errors in the VOR system are caused by multi-path reception. Signals reaching the aircraft receiver may include those that arrive after reflections from natural or man-made objects as well as those arriving by a direct path. The multi-path signals will add and subtract as the phases of direct and reflected signals vary while the aircraft flies along the course.

2) **Ground station errors:** results from misphasing of the 30 Hz reference and variable phases, misalignment of the north and other calibration errors at the VOR station. The major ground station error is due to spurious vertical polarization generated by the antennas resulting in undesirable vertically polarized 30 Hz azimuth dependent component. This spurious 30Hz component will not be in phase with the actual 30 Hz horizontally polarized variable phase. The aircraft antenna, although horizontally polarized, will pick up some of this vertically polarized signal when the aircraft will tilt. These factors may cause an additional error to the tune of $\pm 1^\circ$.

3) **Aircraft receiver error:** It is a function of the cost and age of the aircraft receiver. The older generation aviation receivers tend to have errors, which in new equipment have been essentially eliminated. The modern aircraft receivers have performance better than $\pm 2^\circ$.

4) **Pilotage or flight technical error:** It is a function of many parameters, which are all difficult to measure. Studies of flight technical errors show the error to be higher when the aircraft makes a turn than on a straight-line route.

Since all errors are independent with respect to each other, the total error can be calculated as follows

Total error = $\sqrt{(E_m^2 + E_g^2 + E_a^2 + E_p^2)}$ Where

E_m = Multi-path error

E_g = Ground station error

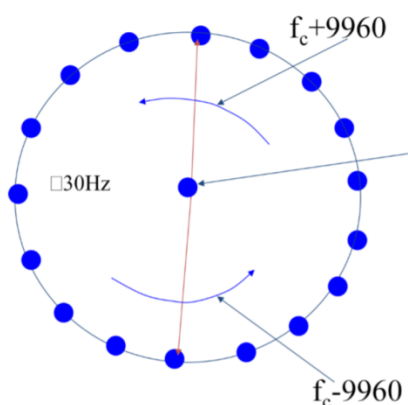
E_a = Aircraft receive

E_p = Pilotage error

3.3.7. Doppler VOR (DVOR)

The DVOR is a later and improved design of VOR which suffers less from siting errors. The CVOR requires a clear area of at least 1500ft in radius. The DVOR is more practical in crowded areas or where there are tall buildings. However, it's a big structure. The DVOR reverses the usage of the two 30Hz signals. However, by also reversing the direction of its rotating variable signal it produces exactly the same result in the receiver. The receiver has no "knowledge" that it's a DVOR as opposed to CVOR it's receiving and operates as

normal. In the DVOR the main VHF carrier is AM modulated at 30Hz - providing the Reference signal. This is transmitted from a central omnidirectional antenna and has the same phase all around the VOR for any receiver. The effect of a 9960 FM modulated subcarrier is created using the Doppler effect by employing a switched array of antenna arranged in a circle of diameter 44ft. (This distance being the exact amount to provide ± 480 apparent frequency shift in the subcarrier.) Imagine a carrier of f_c MHz, AM modulated at 30Hz on the central antenna. Then imagine an array of an even number of aerial elements arranged around the central aerial in a circle of diameter 44ft. (Typically 48 are used.) The VOR controller presents the subcarrier as sidebands on the opposite ends of an imaginary arm. Pairs of opposite aerals are switched in to form a rotating arm at 30Hz (1800RPM). The opposite aerals elements carry are sidebands of $(f_c + 9960\text{Hz})$ and $(f_c - 9960\text{Hz})$ frequencies.



From the receiver's perspective, there's a constant phase 30Hz AM modulation on the main f_c MHz carrier but there also appears to be a 9960Hz subcarrier which is in turn frequency modulated at 30Hz. The sidebands will appear to be frequency modulated at 30Hz by ± 480 Hz due to the rotation and subsequent 44ft variation in distance between transmitting aerial and receiver causing Doppler Shift as the transmitting "arm" rotates. Of course, the phase of the 30Hz frequency modulation on the subcarrier (with respect to the reference signal) will depend on the receiver's angular position around the VOR. Hence, the same receiver comparison will result in the receiver's radial position being established as in the CVOR. The Doppler effect is created by letting the VAR signal be electronically rotate, on the circular placed aerals, at a speed of 30 revolutions per second. With a diameter of the circle of 13.4 meters, the radial velocity of the VAR signal will be 1264 m/s. This will create a Doppler shift, causing the frequency to increase as the signal is rotated towards the observer and reduce as it rotates away with 30 full cycles of frequency variation per second. This results in an effective FM of 30 Hz. A receiver situated at some distance in the radiation field

continuously monitors the transmitter. When certain prescribed deviations are exceeded, either the IDENT is taken off, or the complete transmitter is taken off the air. This receiver is called monitor antenna. It detects the following

- RF level
- 9960 Hz modulation
- 30 Hz AM modulation level
- 30 Hz FM
- Azimuth

The information from the antenna goes to the monitor unit which monitors the error and takes necessary action like shutting down the VOR if the error persists.

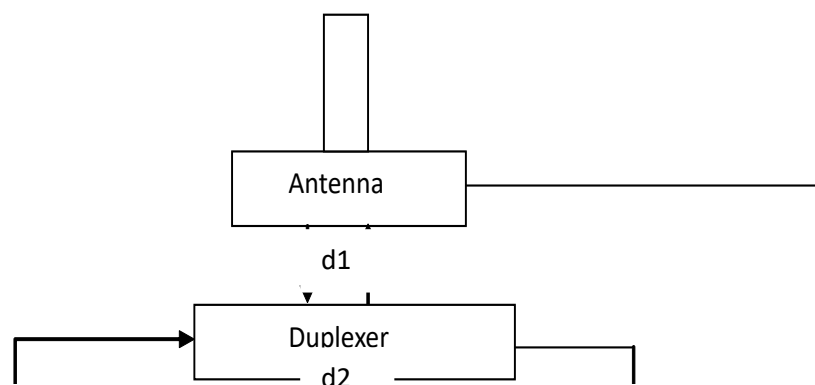
The resulting interference of the REF and the VAR signal gives a pattern known as the cardioid pattern. This cardioid pattern rotates with the VAR signal and covers 360 degrees for one complete switching of the antenna array.

The cardioid pattern is same as the pattern found in the CVOR section above. The receiver antenna on the aircraft receives this pattern and isolates the two signals to detect the azimuth.

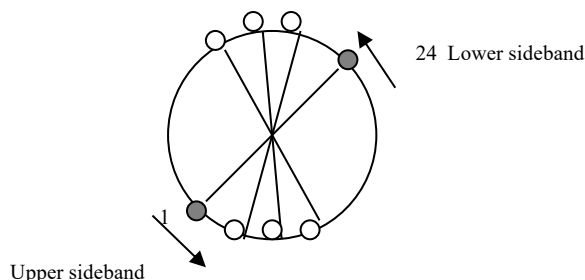
3.3.8. Types of DVOR

Based on antenna rotation techniques, three different types of DVOR have been developed. These are, single sideband, double sideband, and alternate sideband.

Single sideband: - In SSB-DVOR one sideband, $f_c + 9960$ KHz is fed to the commutator and switched around the ring of radiating elements. The system is capable of radiating the correct frequency spectrum. However, in space a receiver finds some variation in field strength. The nearer antenna gives higher field strength than the opposite far end antenna, as they are placed apart. This gives rise to an additional unwanted 30 Hz AM in variable FM signal



Double sideband:- Double sideband operation reduces the above counterpoise effect to almost zero. In this system, upper and lower sideband signals are radiated simultaneously from antennas diametrically opposite to each other. Both sidebands are commuted at 30 Hz in the same direction.



Alternate-sideband:- The alternate-sideband is the simplification of the double-sideband. The two sidebands are radiated alternately from opposite sides. Although technically perfect, it requires some additional modifications in aircraft receiver, and also requires larger counterpoise than the above two.

Out of above three systems, the double sideband VOR became more popular. In Nepal all DVORs are working with double sidebands.

3.3.8 Siting Requirement

The DVOR antenna system consists of an array of antennas on a raised platform in the middle of unpopulated fields. The siting guidelines for a VOR transmitter are given below:

- The site should be on the highest ground in the vicinity to obtain maximum line of sight coverage; preferably 1000 ft – 2000 ft. The land around the station should be circular and as flat as possible. A downward slope up to a gradient of 4% is acceptable.
- The height of the high tension lines or wire fences should not subtend a vertical angle more than 1.5 degree.
- Single trees of moderate size of up to 30 ft. may be allowed beyond 500 ft. No group of trees is allowed within 1000 ft. if they subtend vertical angle greater than 2 degree from the VOR antenna.
- No metallic structure should subtend a vertical angle greater than 1.2 degree or should be within 500 ft. from the station. Wooden structures with negligible metallic contents may be allowed if they subtend vertical angle no more than 2.5 degree.
- In the mountainous terrain, a mountain top site will often be preferable. The site should be on the highest accessible hilltop or mountain, the top of which should be

graded flat to a radius of at least 150 ft. On such sites, the antenna system should be installed approximately a half wavelength above the ground level in the center of the graded area and the transmitter building should be beyond the graded area down the slope below the optical line from the antenna array. No ground trees, power lines, buildings etc. between 150 ft. and 1200ft. should be within the optical line of site of the antenna array.

The VOR system usually coexists with the DME system. The VOR system consists of a central antenna that radiates an omnidirectional reference signal. There are 48 antennas situated at a distance of 7m from the center of the array structure. Each antenna is an ALFORD LOOP ANTENNA. The central antenna is used to continuously radiate REF signals whereas the surrounding antennas form the VAR signal. The signals have been explained in the previous section. The net interaction of the signals gives rise to the rotating cardioids pattern. The counterpoise is a metallic structure on top which is covered by chicken wire. Since the signals from the antennas bounce at the edge of the counterpoise, wider the counterpoise better the immunity from nearby reflections. This causes less multipath reflections and gives space and frequency diversity effects, thereby improving the DVOR. The monitor antenna, placed 80m away from the structure, is approximately 3m from the ground level. It serves to monitor the system and if the signals from the VOR are outside of allowed threshold, the monitor antenna detects this fault of the antenna and transmits the information.

SITING CRITERIA BETWEEN CVOR AND DVOR:-

Obstructions	CVOR	DVOR
1. Flat area without any obstructions	1000-2000'	450'-900'
2. Wooden structures	2.5°	5.6°
3. Group of trees	2°	4.4°
4. Overhead lines	1.5°	3.3°
5. Metallic structures	1.2°	2.6°

6. Fences	0.5°	1.1°
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3.3.9 Advantages of DVOR over CVOR

- When the site is fixed and no alternative sites exist, DVOR gives a higher guarantee of performance and also requires less expertise on site to achieve that performance.
- DVOR has fewer constraints on site selection than CVOR
- In high growth area, the performance of DVOR will not deteriorate with the expansion in the locality
- DVOR is less sensitive to industrial interference.
- DVOR has less maintenance cost, has less frequent flight testing requirements and is more reliable.

The only advantage of CVOR over DVOR is its cost.

3.3.10. VOR receiver system

The primary purpose of the airborne equipment associated with the VOR system is to detect the 30 Hz amplitude modulated signal produced by the rotating cardioids pattern and compare it with the 30 Hz frequency modulated reference. At the output of the 108.0-118.0 MHz receiver is an AM detector. The purpose of this detector is to pick off the various amplitude modulating signals from the VHF carrier.

The detector output is comprised of four elements:

- ❖ Voice modulation, if it has been used at the transmitter
- ❖ Coded 1020 Hz identification tones
- ❖ A 30 Hz signal produced by the rotating cardioids
- ❖ A 9960 Hz tone which has been frequency modulated 480 Hz by the 30 Hz reference signal.

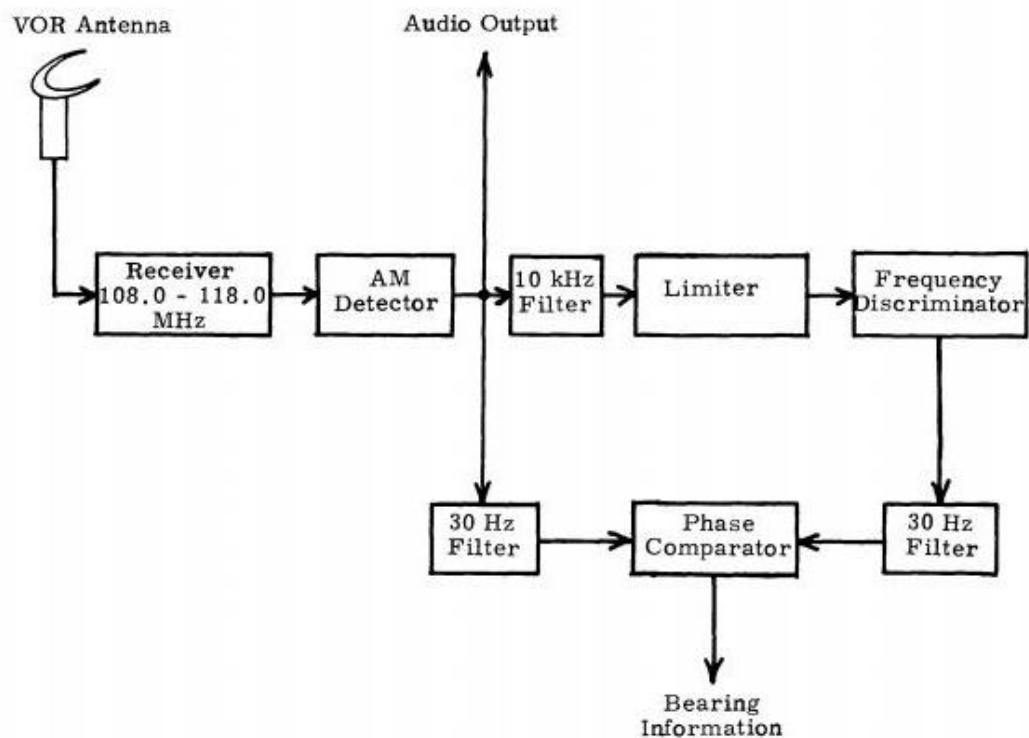


Figure: Schematic Drawing of Conventional VOR Receiver

The voice frequencies and the identification tone are relayed to the audio distribution system of the aircraft. The 30 Hz information which was amplitude modulating the carrier, i. e., the azimuth dependent signal, is filtered to remove other components and fed to the phase comparison circuitry. The 9960 Hz subcarrier information is removed by the 10kHz filter and then limited and applied to an FM detector whose output is the 30 Hz reference signal. After appropriate filtering this is compared with the azimuth dependent signal, and bearing information is the result.

DISTANCE MEASURING EQUIPMENT

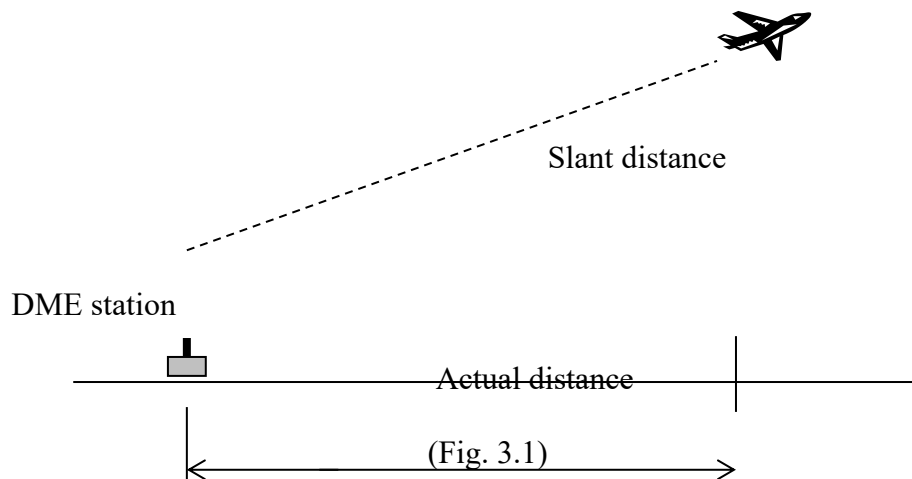
3. DME as a navigational aid

Distance plays a vital role for navigating from one point to other. In aviation, for locating the position of an aircraft polar coordinates (Rho, Theta) system is used, where VOR provides the bearing and DME the distance. Distance Measuring Equipment, or DME, is a standard navigational aid used by all members of the International Civil Aviation Organization (ICAO) for civilian aircraft operation. For military use, a similar system has been developed, which is called Tactical Air Navigation (TACAN). Both operate in the same principal.

Distance measuring is achieved by interactive communication between the aircraft and the ground DME station. For this an aircraft initiates the process by sending a train of paired RF modulated pulses at a rate of 135 pulse pairs per second (pps). Once the aircraft starts getting the replies from the ground DME station the rate is reduced to 27pps. This is called "interrogation" and the aircraft is called "interrogator". After receiving the signal the DME ground station checks the width and spacing of the incoming signal to ascertain that they are within the specified limits. If yes, then it is further delayed to make it exactly 50 us from the time of arrival of the first signal and then responds back to aircraft with a similar pair of signal. This is called "reply" and the ground station is referred to as "Transponder".

Hence DME station provides pilots with a continuous digital display of distance from the aircraft to the facility. Operating on line-of-sight principal, DME furnishes distance information with a very high degree of accuracy. Reliable signals may be received at distances up to 200 NM at line-of-sight altitude with an accuracy of better than ± 0.5 NM or 1.25% of the range, whichever is greater. However, the system is considered to be capable of providing distance information accurate to within ± 370 m (0.2NM) or $\pm 0.25\%$ of slant distance, whichever is greater, for at least 95% of time. Principal factors for maximum range are aircraft height, transmitter power and receiver sensitivity, both in ground and in the air. Distance information received from DME equipment is *Slant Range* distance and not actual horizontal distance.

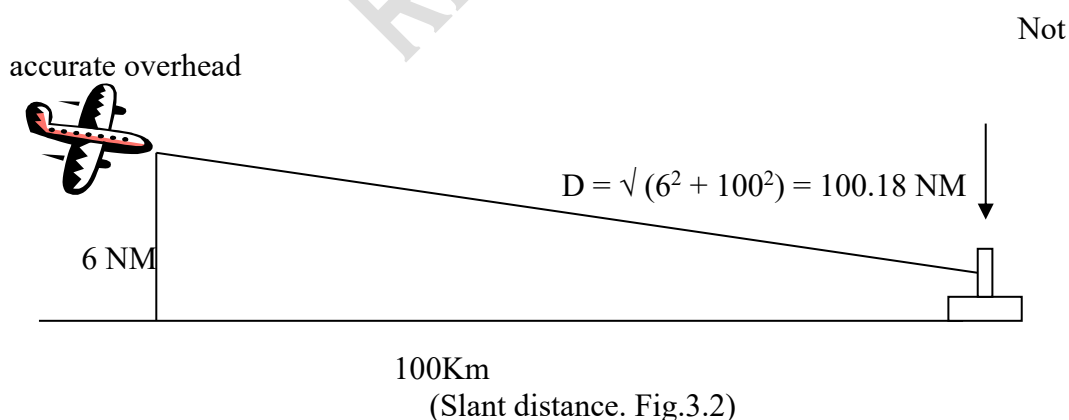
See fig. 1.



Most of the modern commercial jet-aircraft fly below 40,000 ft. (6.59 NM). Therefore, when the aircraft is at a longer distance from the DME station, the slant distance and the actual distance are very close to each other.

3.2. Distance measurement

The distance measured by the aircraft is not the horizontal distance but is the "slant" distance. Since flying height of the aircraft compared to the distance to be measured is relatively very small, there is virtually a very little difference to be counted for. Hence with very negligible margin it could be considered as equal to the horizontal distance. See the following illustration in Fig.2. When the aircraft is 100 NM away from the station and flying at a height of 6NM (36,500 Ft) then the slant distance would be 100.18 NM which is very close to 100NM. Accuracy is higher when the aircraft is far from the station. However, accuracy is lower when closer to the station. Over the DME station, which is the cone of silence, the accuracy is extremely low and cannot be used. However this coverage is small and quickly overflown by the aircraft.



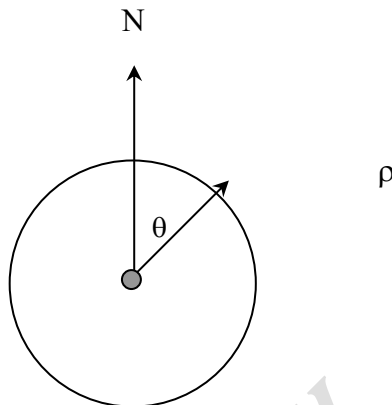
3.3 Applications

Since, DME provides distance information; it can be used in several ways in aviation.

3.3.1 DME co-located with a VOR (VOR/DME) - Rho-theta system

It is the most popular use of DME where a DME is installed together with a VOR. Since VOR provides azimuth and DME distance, they both form together a *Rho-theta* (ρ, θ) system. Thus, an aircraft can find his polar coordinate of any location around the VOR/DME station, which acts as the center of the sphere.

This enables the pilots as well as the ground air-traffic controllers to determine the exact position of the aircraft with respect to the station



(Formation of polar coordinate by VOR/DME. Fig.3.3)

A VOR/DME station can be located at the

- # Vicinity of the runway
- # On the centerline of the runway
- # Or, on the airway routes.

When it is installed around and on the centerline of the runway, an aircraft can use it for homing and departure as well as to align itself to the runway, and make straight-in approach. Such an approach, however, is not very accurate as with an ILS and is called *non-precision approach*. An ILS approach is fully reliable hence it is called *precision approach*. In the places where an ILS is not available, non-precision approach is very helpful. Kathmandu airport uses DVOR/DME non-precision approach for all landings and takeoffs day and night. When a VOR/DME station is located away from the airport, it is mostly used for en-route aid, which provides position fix and route guidance. The illustration in Fig. 2.3-3 shows how an aircraft can make turn at points A and B using VOR/DME systems.

In most of the VOR/DME installations both the equipment are placed inside the same shelter and the DME antenna is located on the same vertical axis of the VOR antenna. It is called *coaxial collocation*. In DVOR/DME installations, however, due to space restrictions the DME antenna may be installed as far as 80m from the VOR antenna system. In some other situations, the antenna separation may be much higher, but in any case it should not exceed 600m (2000 ft.). When antennas are separate, it is called *offset collocation*.

In VOR/DME installations, DME frequency is paired with VOR as per allocations made in ICAO Annex-10. Therefore, as soon as a pilot tunes to the specific VOR frequency, if the DME is collocated, it is received automatically. To identify collocation, both VOR and DME share the same station identification code. The identification code is repeated seven times per minute, with three times for VOR and once for DME, and so on. In such installations, the range of DME should be the same as VOR. The radiation pattern of both equipment is Omnidirectional. If VOR is not available, DME is sometimes collocated with a NDB. It serves the same purpose but with less accuracy with respect to azimuth guidance. In NDB/DME installations the DME frequency is not paired. Therefore, both have to be selected independently. In Pokhara NDB/DME collocation has been provided.

3.3.2 DME with ILS (ILS/DME)

At some geographical locations, where installation of associated Marker Beacons of the Instrument Landing System (ILS) is not possible, a DME can be installed to provide the distance information. When DME is used as an alternate to Markers, the DME is located on the airport and adjusted in such a way that the zero range indication will be a point near the end of the runway. Also, to reduce the angular error the DME antenna should not be more than $\pm 20^\circ$ from the centerline of the runway. In most of the cases, the DME is normally located inside the Glide Slope (GS) shelter of ILS. The Glide Slope station is normally 250 to 300 meters from the end of the runway, and is only offset from the centerline by 120-150 meters. Therefore, it meets the above requirements. See fig. 2.3-4. In ILS/DME installations, DME frequency is paired with Localizer frequency and they both share the same identification tone, like with VOR. While it is not specifically required that DME be frequency paired with the Localizer, in most of the cases when it is used as an alternate to Outer Marker, frequency pairing is preferred to simplify pilot operation.

Where only Localizer service is provided, it can be collocated with the Localizer. DME is also installed with the Microwave Landing System (MLS), which is an alternate to ILS with better accuracy and ideal for difficult sites. DME collocated with an ILS or MLS system should have directional radiation pattern with distance accuracy better than ± 0.2 NM.

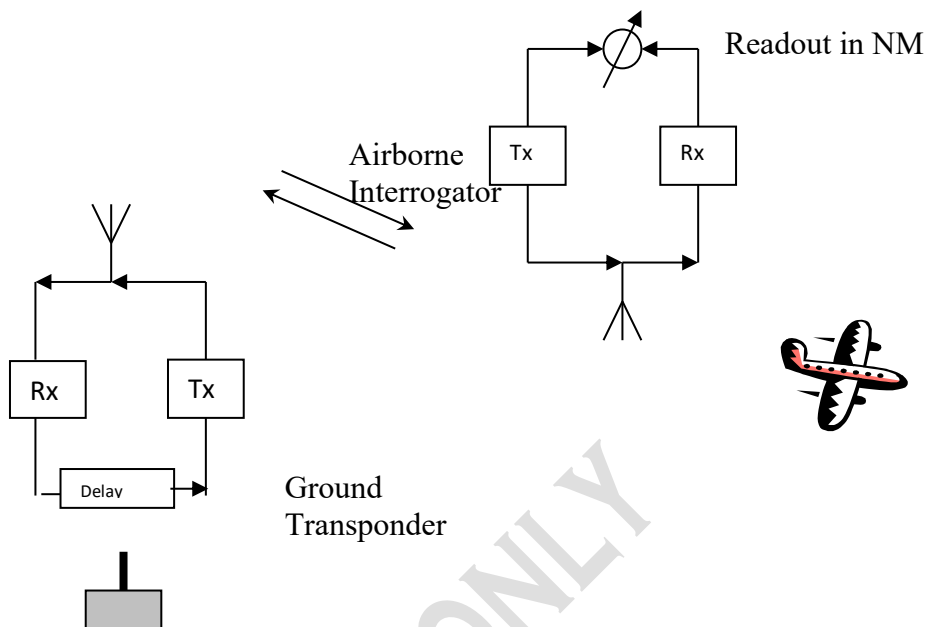
3.3.3 DME alone

DME is also installed as an independent station. In such installations the radiation pattern is normally omnidirectional.

3.4 Principal of operation

DME ground system, which has transmitter/receiver, called *transponder*, works in conjunction with airborne transmitter/receiver, called *interrogator*. The principal is that the interrogator transmits continuously a series of interrogation pulse pairs to the

transponder, which are received by the *transponder* receiver. After checking the correctness of the incoming pulses the transponder holds for specified delay period and transmits back the reply pulses. The time difference between interrogations and reply pulses are measured in the interrogator receiver which is computed into distance information to display directly in nautical miles. In navigation the distance is always measured in nautical mile. 1' (minute) latitude or longitude represents 1NM. (1 NM = 1.86 KM).



(Distance Measuring System. Fig.3.6)

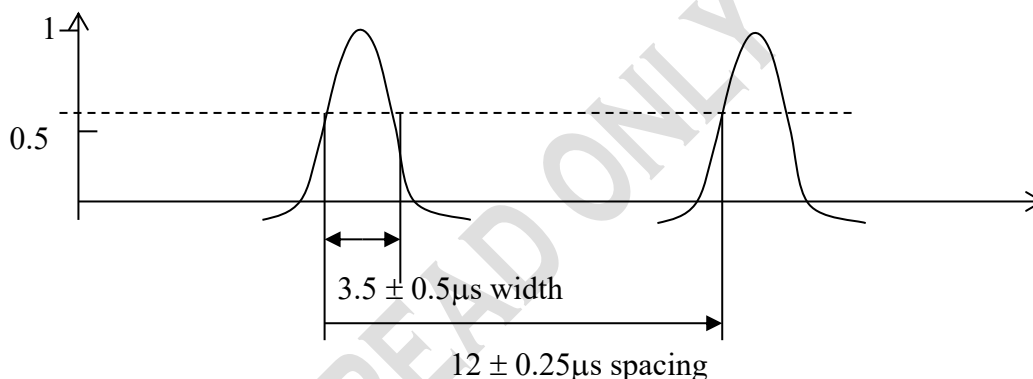
DME operates in UHF frequency band from 960 MHz to 1215 MHz. The band is divided into 126 1-MHz channels for interrogations, and another 126 1-MHz channels for replies. There is always a difference of ± 63 MHz between interrogation and reply pulses. When DME transponder is intended to operate with an ILS, VOR or Microwave Landing system (MLS), its frequencies are paired with associated navigation system. The details of these channel pairing is indicated in ICAO Annex-10. Thus, a pilot only tunes to ILS; VOR or MLS frequency channels and receives automatically the distance information when a DME is collocated with any of them. To identify a particular station, DME transmits identification codes at a fixed repetition rate, which varies in accordance with installations. If alone then it is at the rate of 6 words per minute.

For obtaining the distance information, it is just required to tune the VOR frequency which will then automatically tune the DME Frequency as they are paired with each other. The aircraft interrogator starts transmitting a series of double pulses at a *Pulse Repetition Frequency (PRF)* of 135 Pulse Pairs per Second (PPS). It is called the *Search Mode*. For modern interrogator equipment the search time is just 1-2 seconds. As soon as the interrogator receives the reply signals from the ground station PRF decreases from 135 PPS to just 27 PPS and starts displaying the distance information. It is called *Track Mode* or *Lock Mode*. A modern DME station is capable of proving up to 2700 PPS.

Therefore, a maximum of 100 aircraft may receive distance information simultaneously from one transponder. However, 95 aircraft will be in lock mode and 5 in track mode. When there is no interrogation from the aircraft, the transponder receiver generates the interrogation signals internally at the rate of 2700 PPS and receives the reply in order to keep on activating the transponder continuously, and to monitor the performance of the station. As the aircraft interrogations are increased the internal interrogations are automatically decreased at the same rate to keep overall PRF to 2700.

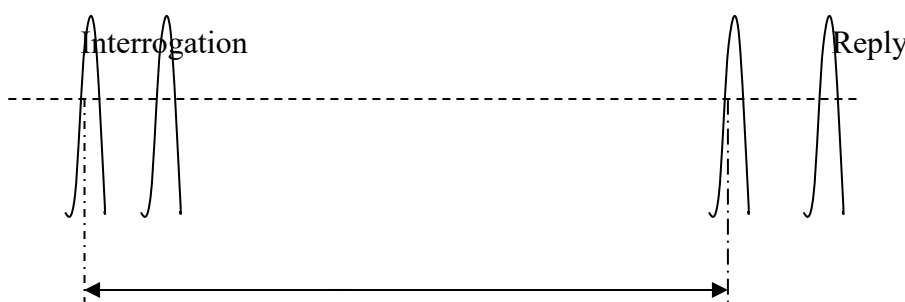
For interrogation as well as for reply DME uses a pair of pulses, called *Gaussian Pulses*, which are $12 \pm 0.25 \mu\text{s}$ apart and $3.5 \pm 0.5 \mu\text{s}$ wide. The frequencies of interrogation and reply, however, differ by $\pm 63 \text{ MHz}$ from each other.

After receiving a pair of interrogation pulses the DME receiver checks the width and spacing of the pulses, holds it for a total of $50 \pm 1 \mu\text{s}$ and then triggers back a reply. Therefore, from the start time of the reception of the pulses the transponder receiver would not accept any new incoming signals for $50 \pm 1 \mu\text{s}$. This is called “Receiver Dead Time”..



(Pulse width and spacing of interrogation and reply for DME. Fig 3.7)

The DME receiver dead time of $50 \pm 1 \mu\text{s}$ is necessary to make all the DME equipment similar in performance as the actual circuit delay could vary from 20 to $30 \mu\text{s}$ from equipment to equipment that may lead to unacceptable errors. Furthermore it has significant importance in echo suppression. This would be dealt with in detail in the following paragraphs.



$$50 \pm 1\mu\text{s}$$

(Receiver delay or dead time. Fig.3.8)

3.5 Gaussian Pulse

The DME system uses Gaussian Pulses instead of rectangular pulses, as normally is in the case of primary Radar system. The reason for this is that the DME channels are very closely spaced, i.e. 1-MHz apart.

If rectangular pulses were used then the frequency spectrum would follow a $\text{Sin}X/X$ form and the energy would spread outside the 1-MHz channel bandwidth. This would cause the energy to pass into adjacent channels, which may give rise to unnecessary interference in co-channels.

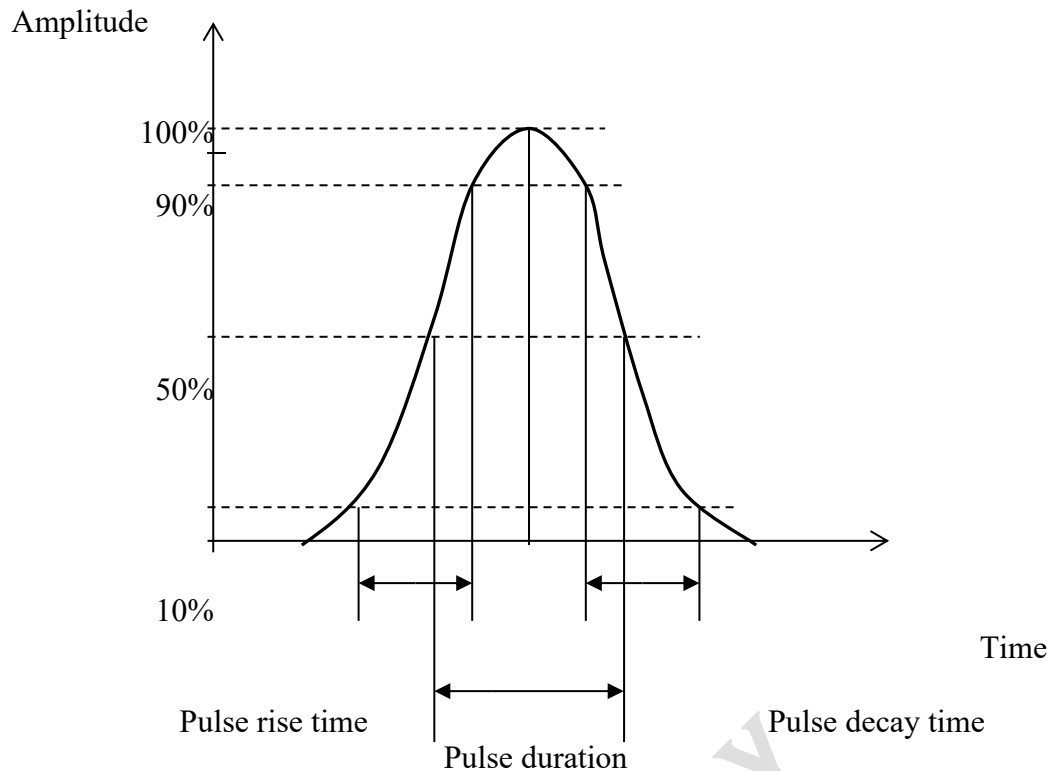
To decrease the spectrum width it is necessary to reduce harmonics in the pulse. That's why the Gaussian pulse has been chosen. Mathematically it can be proved that a Gaussian pulse has relatively smaller frequency spectrum. Hence, most of the energy can be maintained within the 1-MHz channel and interference with co-channel stations is reduced. The Gaussian pulse can be represented by the formula:

$$f(x) = Ae^{(-t/\sigma)^2}$$

Where "A" is the amplitude and "σ" is the pulse half duration at 1/e point.

But this is at the expense of accuracy in distance indication, because if the detection level will vary (which normally would occur due to shape of the pulse), it will result in variation of time. In DME, the measurement of time is done at half amplitude of the Gaussian pulse. Therefore, any distortion in shape may cause distance error. A variation of just 1 μs may cause an error of approximately 150 meters.

ICAO Annex-10 specifies the shape of the Gaussian pulse. The Gaussian pulse has been illustrated in Fig.9. To reduce the harmonics and the distance error, the pulse should be obtained in the equipment as accurately as possible.

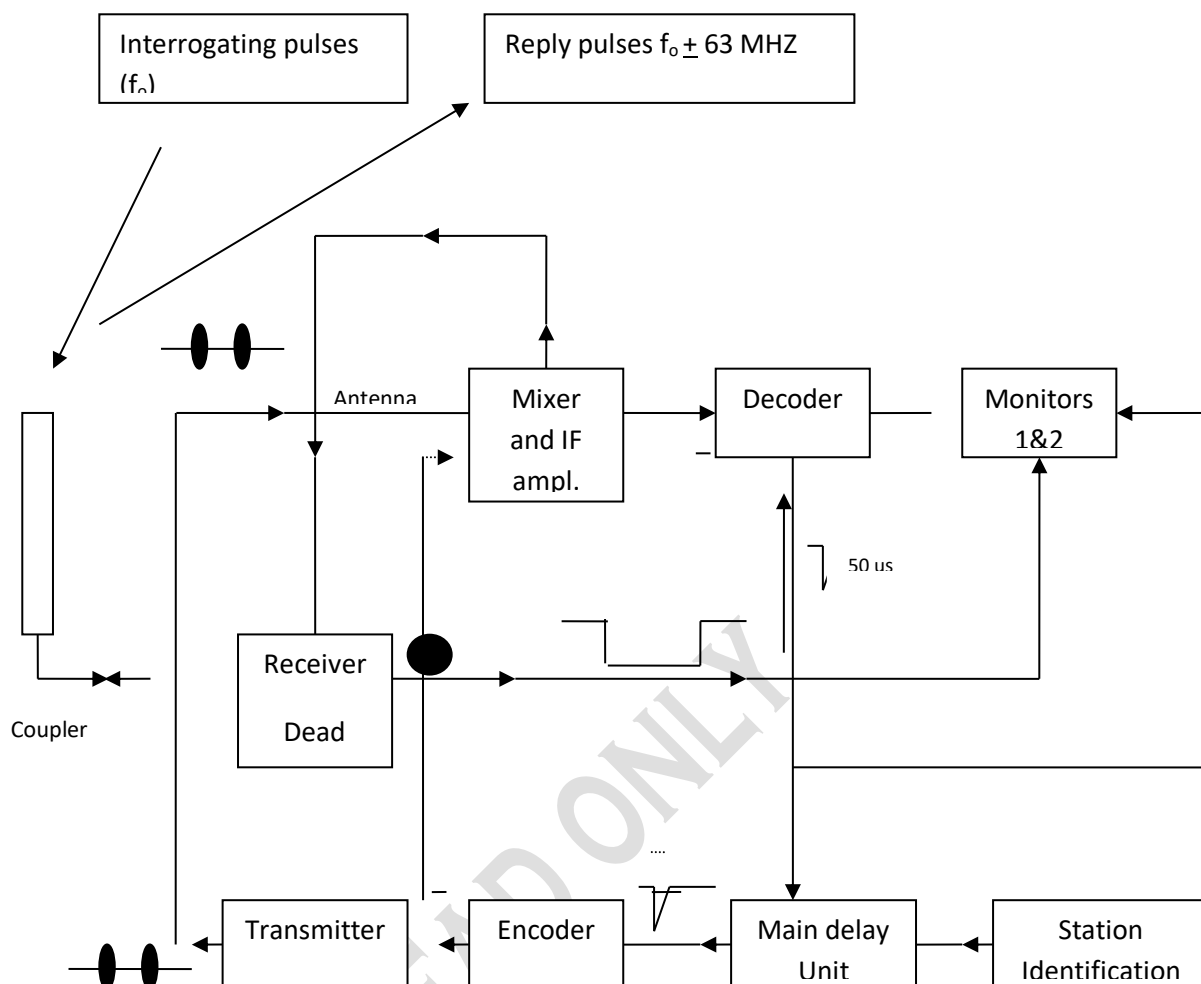


(Gaussian pulse : Fig.-3.9)

- Pulse rise time - The time as measured between 10 and 90 per cent amplitude points on the leading edge of the pulse envelope. (2.5 - 3 μ s)
- Pulse decay time-The time as measured between 90 and 10 per cent amplitude points on the trailing edge of the pulse envelope. (2.5 - 3 μ s)
- Pulse duration - The time interval between the 50% amplitude point on leading and trailing edges of the pulse envelope. (3.5 $\pm$ 0.5 μ s)

3.6 DME Transponder Operation

A simplified bloc diagram for a general DME transponder is shown in Fig. 10. The transponder antenna, which is normally a stacked array of conical dipoles, receives interrogation pulses. Polarization of antenna is vertical and it radiates omni-directionally in the horizontal plain with 9dB gain at 3 degree over the horizon. The antenna works iL-band.



(Simplified Bloc Diagram of Transponder. Fig. 3.10)

The coupler isolates the Receiver and Transmitter signals and hence the Interrogating pulses are passed to the Mixer, which gives 63 MHz (difference between interrogating and reply frequencies).

The signal is amplified in the IF unit and also passes through a Ferris Discriminator which is a very high selective Band Pass filter. Normally in the IF unit the signal is also mixed down to a 2nd IF frequency around 11 MHz. the signal is then detected and passed to a Decoder which checks that the pulse spacing of the so called video pulses (LF pulses) are within 12 ± 1 μ s. If so, the Decoder triggers a short spike pulse with reference to the 2nd pulse in the pair.

Normally the total **system delay** in a modern transponder circuitry is approximately 20 μ s (including the 12 μ s delay in decoder) hence the Main delay circuit must delay the pulse spike for further 30 μ s to obtain the over all delay of 50 μ s. The Main delay circuit is mainly a simple Monostable multivibrator. The Coder (or Encoder) will for each spike input give out a double Gaussian pulse pair with the correct pulse spacing and puls

characteristics given by ICAO. (The Coder consists mainly of multivibrators and a Gaussian filter). This video signal modulates the Transmitter, which produces RF pulse pairs with correct frequency. The Transmitters are either Low power (100 Wp transistorized PA) or Medium power (1kWp PA including valves) or High power (5kWp with klystrons). Here Wp denotes “pulse power” that is much lower than the average power of the transmitter. Pulse power is the power of the transmitter for a very short period while transmitting the particular pulse. The frequency is always 63 MHz above or below the correct interrogating frequencies.

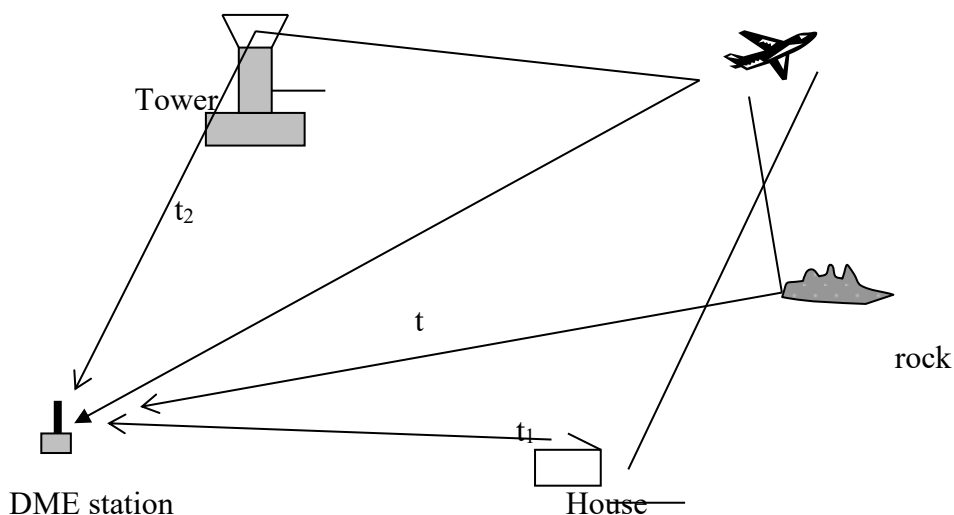
The Transponder also transmits identifications signal at every 30 seconds when co-located with VOR or ILS-LLZ. The identification signal has the frequency 1350 pulse per second and do have 2 or 3 letters in the Morse code which indicates the signature of the ground beacon.

Even if no aircraft is present to interrogate the transponder the duty cycle must be kept constant 2700 pulse pairs and this is carried out by the Monitor, which gives noise or squitter pulses inversely proportional to interrogating pulses. Therefore, with no interrogations all 2700 pulse pairs will come from the Monitor. On the other hand, with 100 aircrafts interrogating, there will be no pulses coming out from the Monitor, because all 2700 pulse pairs will be produced by the aircraft.

3.8 DME errors and echo suppression

3.8.1 DME errors

DME works in UHF band, therefore, strict line-of-sight principal applies to it. DME mainly suffers from *multi-path error*. Since DME antenna in the aircraft is not directional, the interrogation pulses from the aircraft may also be reflected from the surrounding terrain; buildings etc. and arrive later as the *echo pulses*. See Fig. 12. The echo pulses, if they are strong and within the specified limits (i.e. correct width and spacing), they may also be accepted by the transponder as the true signals. Consequently, false replies may be triggered back. These replies originating from echo pulses could be accepted in some aircraft receivers and may cause false indic

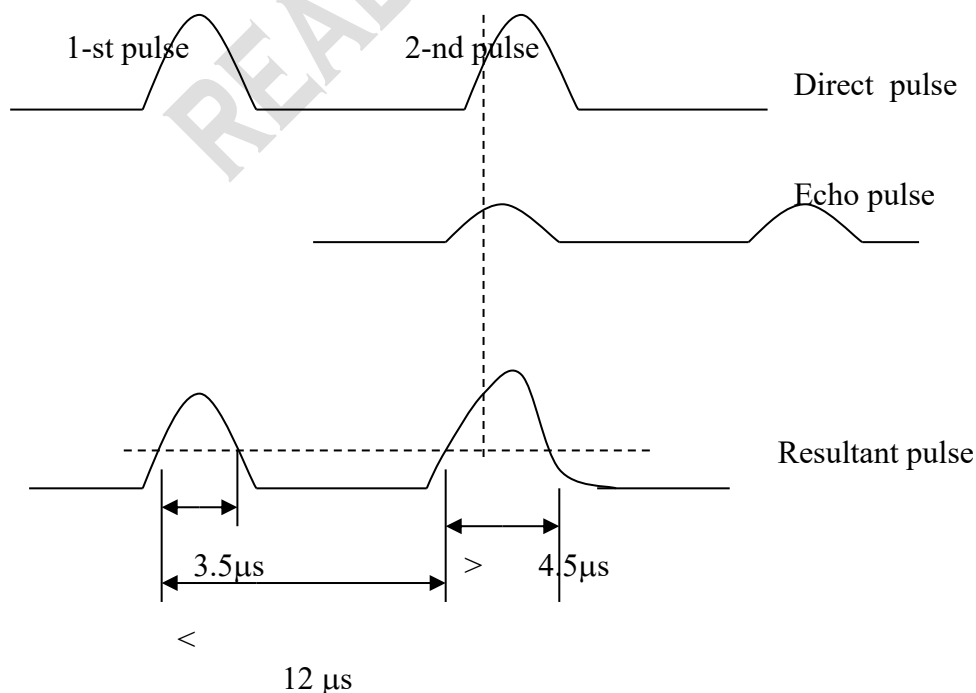


3.8.2 Echo suppression by DME dead time

To eliminate echo to some extent, *DME dead time* is very useful. The DME dead time is a period of blanking of the transponder receiver during which no incoming signal is accepted. In most of the DME transponders the dead time is adjusted to $50 \pm 1 \mu\text{s}$. If the reflecting points are within 5 NM from the DME station then most of the echo pulses will be rejected by the DME receiver. However, the long distance echo pulses, if they are strong, may cause problem. The following illustration clarifies the above statements.

3.8.2.1 Short distance echo

The echo pulses may arrive in phase or out of phase compared to direct pulses. The Fig.13 illustrates the situation when both pulses arrive in phase. If the first pair of the echo pulse arrives with a delay of, say, $10 \mu\text{s}$ then the first direct pulse will not be distorted. However, the second direct pulse will add up with the first echo pulse. From the above it is seen that after addition the width of the second pulse gets wider. If it is more than $4 \mu\text{s}$ then the DME receiver will reject the pair. Similarly, when the echo pulse will arrive anti-phase then the composite waveform will be less than $3.5 \mu\text{s}$, which will again be rejected by the receiver. To avoid this situation blanking of receiver for some time is necessary, which is referred to as DME dead time or receiver dead time. During this period no other pair is accepted until a reply has been made in response to that particular pair. If the receiver dead time was not there any echo pulse that will arrive during that period would have been accepted by the receiver. This would have created either rejection of valid pulses due to signal deformations or false distance indications due to echoes



(Deformation of pulse pair due to echo. Fig. 3.13)

Receiver dead time is normally adjusted to 50 - 60 μ s. It will protect from echo signals that will generate from reflections closer to DME station (up to 5 NM). These are the short distance echoes.

3.8.2.2 Long-distance echo

Long-distance echoes are those which arrive after the receiver dead time. Normally the long distance echoes are weaker. Therefore, they are below the receiver threshold point and rejected by the transponder. However, sometimes the far distance echoes may be strong enough to be accepted by the receiver and trigger the replies causing false lock on problem. To avoid the situations the receiver dead time may be increased further more than the normal 50 to 60 μ s.

By increasing the receiver dead time *false lock on* problem may be reduced but this will affect on overall *reply efficiency* of the Ground station. This is because during the dead time the transponder receiver will reject all the incoming signals from other aircraft. *Reply efficiency* is a factor that indicates the ability of the transponder receiver to receive interrogations and make replies successfully. There is a relationship between efficiency and dead time.

Reply efficiency = 1 - 2700X receiver dead time.

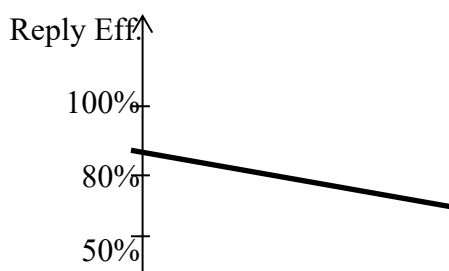
For 50 μ s receiver dead time we get:

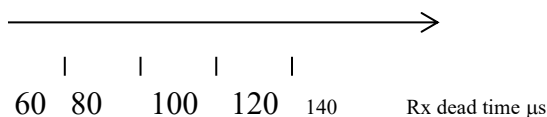
Reply efficiency = $1 - 2700 \times 50 \cdot 10^{-6} = 0.865$ (86.5%)

For 100 μ s receiver dead time we get:

Reply efficiency = $1 - 2700 \times 100 \cdot 10^{-6} = 0.73$ (73.5%)

Thus, by increasing the receiver dead time while we can suppress the long distance echoes, we reduce the reply efficiency of the ground system. So length of the receiver dead time should be taken in to consideration only after examining the nature of the echoes. The following illustration in Fig. 14 shows the relationship between receiver dead time and reply efficiency of the DME. If the dead time is more than 150 μ s then the reply efficiency in practice will be 50%, which is the lower threshold of an aircraft interrogator to maintain the distance information. ICAO recommends to keep the dead time not exceeding $60 \pm 1 \mu$ s unless the long distance echoes are too prominent to be neglected. Even then it should be increased only by the minimum amount just necessary to allow the suppression of echoes



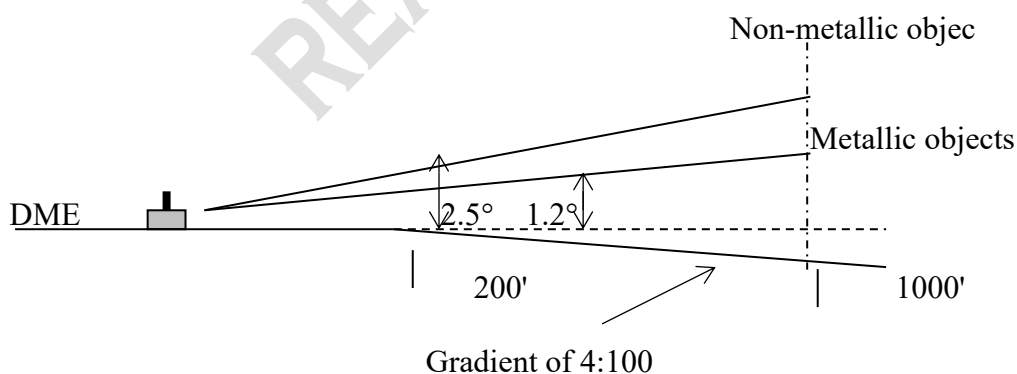


(Receiver dead time vs. reply efficiency, Fig.3.14)

Another factor, that affects the reply efficiency, is the receiver sensitivity or receiver threshold. In order to accept most of the aircraft signals the receiver sensitivity of the ground equipment should be very high. In any case, if the incoming pulse pair strength is -120 dBW/m^2 the transponder will reply with an efficiency of 70% or more. The transponder output power is normally kept at 1KW pulse peak power.

3.9 Siting requirements

The basic requirements in siting a DME beacon are to ensure adequate coverage and to avoid the possibility of interference to the correct operation of the aid. Site selected in open country should keep hills, mountains, large buildings, etc. at as small angle of elevation as practicable. The Fig. 15 shows the basic site requirements of a DME station.



(Basic site requirement of a DME, Fig. 3.15)

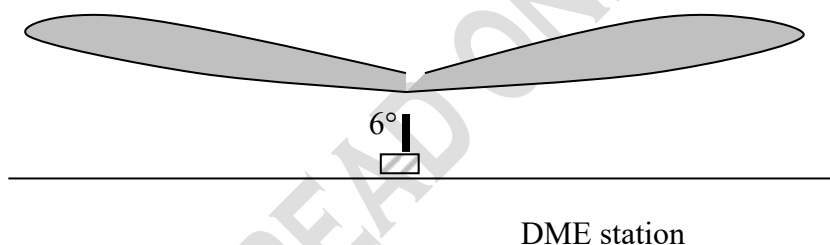
The distant obstacle horizon should preferably not extend above an elevation angle of 0.5° when viewed from the near ground level at the proposed location of the DME. Within 200' from the DME antenna the area should be flat and clear of all obstructions. No group of trees or overhead lines are permitted within this radius. Beyond 200' a downward slope of 4:100 is permitted. Within 200' - 1000' from the DME all metallic

objects should not subtend an angle greater than 1.2° . For non-metallic obstructions up to 2.5° is allowed. As a general guidance, small buildings, power and telephone lines and fences can be tolerated within 200' provided they are not higher than the DME antenna. Normally a DME antenna is kept up to a height of 20' from the ground if that clears local obstructions.

Large buildings such as multi-story buildings, steel bridges, metallic towers etc. are potential sources of interference. If they are within 3 NM from the station they may cause signal deformations. All the houses within 1000' should be constructed lengthwise and along the radials from the DME station as far as practicable. DME is highly affected by electrical noise. Therefore, any high-tension line above 22KV should be kept as far as 3000'. There are no restrictions on vehicular movements around the site.

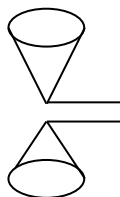
3.10 Antenna system

Since DME suffers from echo signals generated by multi-path effect, highly directional antenna system is used to avoid unwanted reflections. The DME signal is vertically polarized. In non-directional stations, such as in VOR/DME, the radiated signal is Omnidirectional with slightly tilted beam width of approximately 6° . This provides desired power on the horizon necessary for minimum echo generation. See fig.16.



(Radiation pattern of DME. Fig. 3.16)

To achieve such a low beam width stacked biconical antenna radiating elements are used. They form together an antenna array, which provides narrow radiation pattern of 6° . The difference between maximum and minimum azimuth points is not more than 2db.



(Biconical antenna element. Fig. 3.17)

When a DME is installed with an ILS highly directional antenna system is used. Furthermore, in such an installation the transponder time delay is adjusted in such a manner that the aircraft interrogator indicates zero range at a specified point.

3.11 Monitoring and calibrations

The DME is a highly accurate and dependable aid, which provides distance information to the aircraft. Therefore, the independent monitor units constantly monitor its performance. Normally up to two monitors are used. In the event that any of the conditions specified below occur, the monitors will cause the following actions to take place:

- # a suitable indication shall be given at the aircraft cockpit.
- # the operating transponder shall be automatically switched off and the standby transponder will be turned on.
- # The monitors continuously measure the following radiated parameters of the DME:
 - # a fall of 3db or more in transmitted power output.
 - # Pulse spacing of 12 μ s exceeds more than $\pm 1\mu$ s
 - # Reply delay exceeds by $\pm 1\mu$ s
 - # Reply efficiency < 70%
 - # Identification tone not repeated every 30 seconds or transmitted continuously for more than 5 seconds.
 - # Pulse counts < 850 pulse pairs per second.

Monitoring signals are obtained from the pick up probes closely placed near the antenna elements.

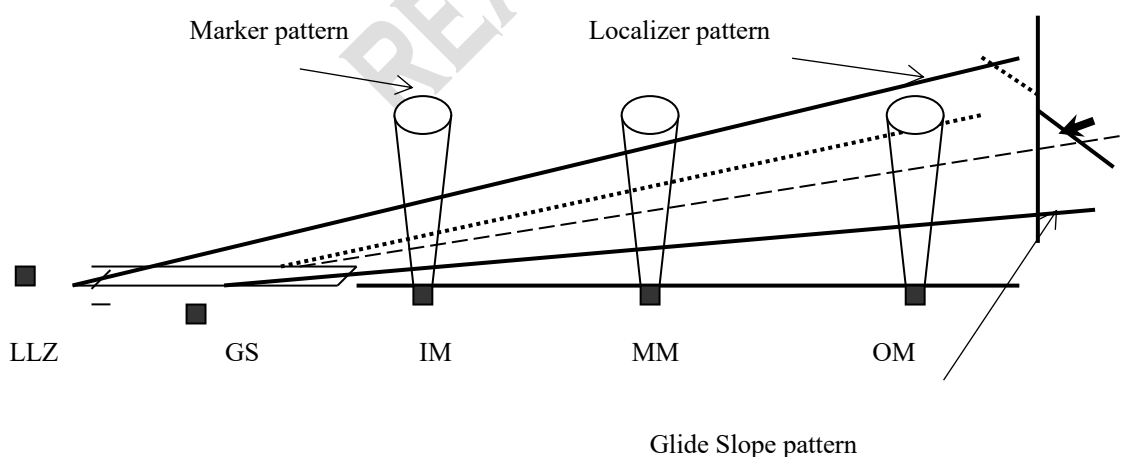
Like in other navigational aid equipment, calibration is done in regular intervals, both in the ground and air. While ground calibration is carried out by using specific measuring test equipment, for the flight calibration specially equipped aircraft is deployed. The aircraft normally checks the DME coverage area, field strength, reply efficiency and echoes in the specified routes and places.

Instrument Landing System

The Instrument Landing System, abbreviated as ILS, is a system of electronic equipment, which assists the landing aircraft to make straight in approach by using cockpit indications at any non-visual meteorological conditions. ILS is a standard aid, adopted by the members of ICAO since its development in 1940's. There are hundreds of ILS's in operation at all modern airports throughout the world. It is still considered to be the most reliable, most utilized and most implemented precision approach system in the world.

The system comprises of a **Localizer**, a **Glide Slope**, and two to three **Marker Beacons**. The landing path is determined by the intersections of two planes, as shown in Fig.1-A, and could be explained as follows:

- A vertical plane containing the runway centerline, is defined by a VHF Transmitter, called Localizer (LLZ).
- A horizontal plane of 2°-4° vertical angle containing the runway centerline, is defined by an UHF transmitter, called Glide Slope (GS).
- Vertically radiated VHF Markers (IM,MM & OM) transmitters provide fixed distance information
-



(Radiation pattern of an ILS)

All these stations form a system that provides an electronic passage, exactly at an approach angle that is required for a safe landing.

The ILS helps to bring the aircraft safely down to a pre-defined height, called the Decision Height, from where the pilot has to make his own decision whether to land or to make a missed approach. The missed approach is an aviation terminology for unsuccessful landing. In this case, the aircraft has to make a turn and try to land once again. In category- IIIC, visibility is not needed and a blind landing can be made using electronic equipment. Therefore, based on *decision height* and the *visibility* of the runway, three categories of ILS are defined by the International Civil Aviation Organization (ICAO) which is tabulated below.

Table -T1

ILS CATEGORIES	DECISION HEIGHT (M)	VISIBILITY (M)
CAT - I	60	800
CAT - II	30	400
CAT - IIIA	0	200
CAT - IIIB	0	50
CAT - IIIC	0	0

Since the pilots fully rely on ILS guidance for landing, the signals radiated by an ILS should be very accurate and authentic. ICAO Annex-10 specifies the necessary technical tolerances that have to be maintained for the above three categories of ILS's.

In many occasions, where the geographical conditions do not permit installation of Marker beacons at predefined distances, Distance Measuring Equipment (DME) is co-located with ILS. The DME can be co-located with an ILS in the following three ways

- DME with Glide Slope
- DME with Localizer
- DME installed independently.

Thus, while landing on ILS, a pilot determines his position from the runway end by DME. In some locations, an NDB is installed on the centerline of the runway in stead of a Marker. Such an NDB is then called a compass locator.

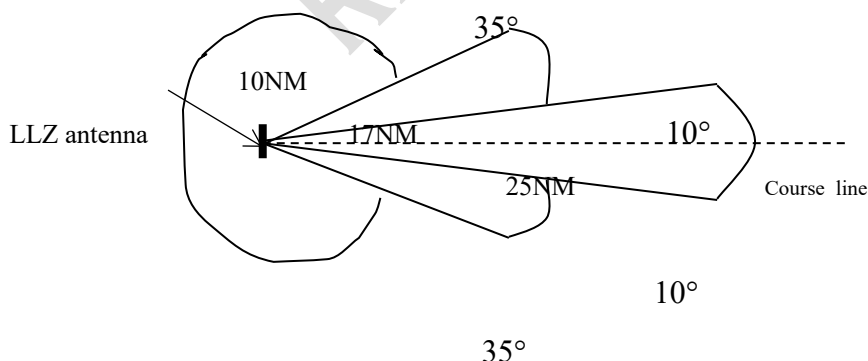
Coverage of an ILS

Localizer: The horizontal localizer coverage sector is extended from the center of the localizer antenna array to the distances of:

46.3 km (25 NM) with $\pm 10^\circ$ from the from course line.# 31.5 km (17 NM) between 10° and 35° from the front course line.

18.5 km (10 NM) outside of $\pm 35^\circ$, only if the coverage is needed.

Here, the course line means the extended centerline of the runway. In most cases the coverage is limited to $\pm 35^\circ$ only. Where the topographical features do not permit a longer range, the localizer radiation can be reduced to 18 NM instead of 25 NM within $\pm 10^\circ$, and 10 NM instead of 17 NM between 10° and 35° lines.



(Horizontal coverage of localizer)

Fig.4.2

Difference in Depth of Modulation

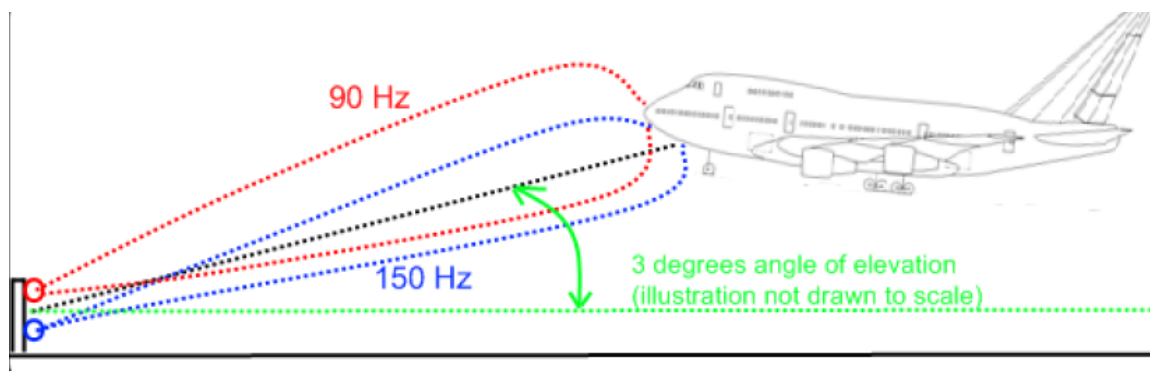


Figure: Difference in Depth of Modulation for ILS-GS (side view)

The total beam width is approximately 1.4 degrees and the angle of the desired approach glide slope is 3 degrees. By demodulating the received glide slope signal and calculating the difference in depth of modulation (DDM) between the two tones, the ILS-GS provides the pilot with vertical course data. Difference in depth of modulation (DDM) is the relative difference between the two AM carriers of 90 Hz and 150 Hz. If the DDM is a positive value, the upper 90 Hz beam is predominant, whereas if the DDM is a negative value, the lower 150 Hz beam is predominant. A pilot would need to exercise motor judgment to maintain enough power and approach angle for ideal vertical descent (DDM=0). The landing approach angle must also be corrected so that the aircraft landing will be “cushioned” by air upon touchdown and be within tolerances of the landing gear’s mechanical integrity.

4.2.2. Marker Beacons:

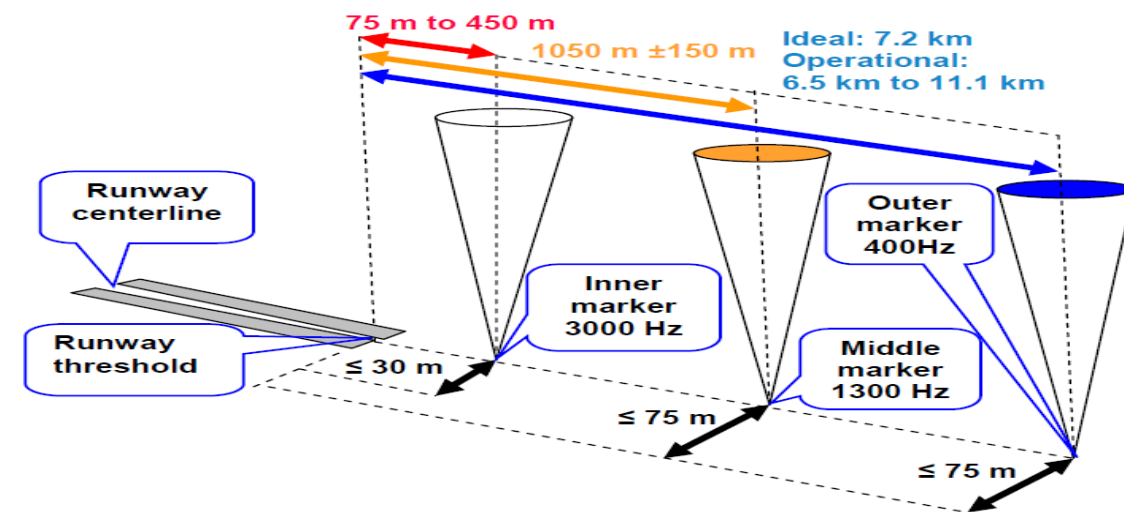


Fig: Marker Beacon Placement with respect to runway

Normally, only two Marker beacons are installed in most of the locations. These are, Middle Marker (MM) and Outer Marker (OM). The Inner Marker (IM) may be added whenever its need may be felt at any particular site.

All the Markers radiate vertically in an elliptical shape on the course line through which the aircraft makes the approach. The Marker beacon system should be adjusted to provide the coverage over the following distances, measured on localizer and glide slope intersection:

a) *Inner Marker*: $150 \pm 50\text{m}$ ($500\text{ft} \pm 160\text{ft}$)

b) *Middle Marker*: $300 \pm 100\text{m}$ ($1000\text{ft} \pm 325\text{ft}$)

c) *Outer Marker*: $600 \pm 200\text{m}$ ($2000\text{ft} \pm 650\text{ft}$)

4.2.3. Siting Requirements

Since ILS transmitters work on mid-VHF band and they carry sensitive navigation signals; care must be taken for its proper siting requirements. The ability of an ILS system to provide a reliable signal depends upon proper formulation of radiation pattern, and absence of natural and man made objects that may cause signal aberrations.

In the early days, installation of an ILS was a major problem restricting its use only to the airports with clear and plain terrain with little or no man made obstructions in the vicinity. Nowadays, due to advancements in technologies, companies developing and manufacturing ILS have improved the design and performance to such an extent that an ILS can be installed even at a very difficult location. Especially, in the last one decade ILS technology has progressed to include important innovations in the field of antenna and microchips, which made it adaptable to any siting challenge and more reliable than the old systems.

Since the quality of the radiated signal is highly dependent on topographical features surrounding the airports, the problems are successfully overcome these days by utilizing special antenna configurations, and the manner in which RF signals are fed to them.

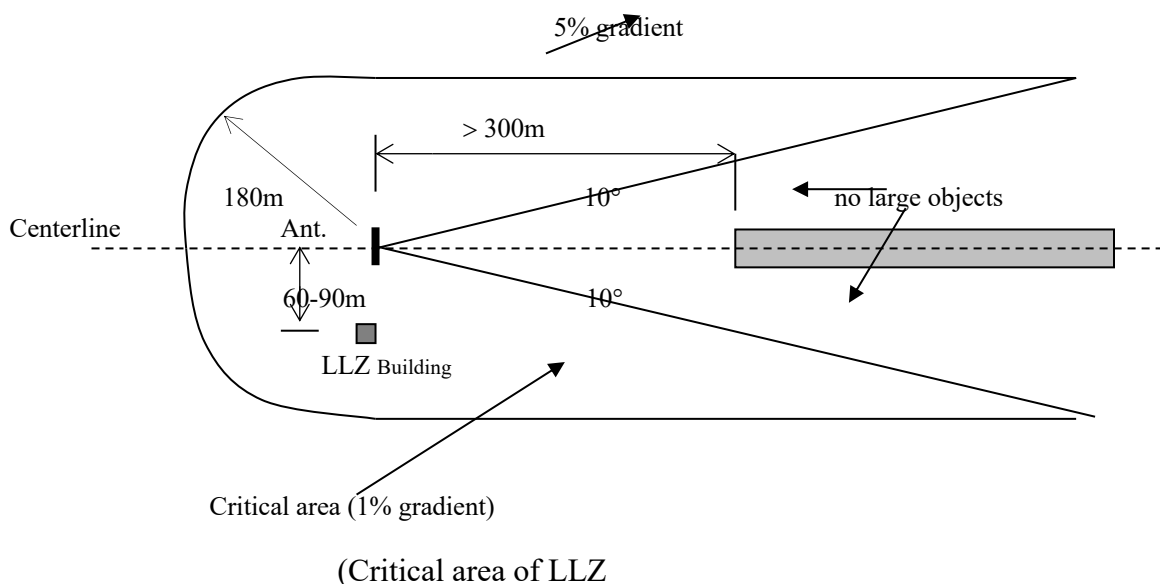
Therefore installation may vary significantly from site to site. As a general rule, the following siting criterion is adopted.

Localizer siting criteria

The preferred location for a Localizer is on the centerline of the runway beyond 300 meters from the stop end. The distance between the end of the runway and antenna positions may be varied to suit a particular condition. The antenna array could even be located beyond 600 meters to allow for planned runway extension. However, it will reduce the course width of the Localizer. On the other hand, it is recommended not to reduce the distance by less than 300 meters, as antenna might be subject to high intensity jet propulsion, and at busier airports field measurements may have to be restricted.

The shelter containing electronic equipment is generally located at 60 to 90 meters to one side of the antenna system. The shelter could be located behind the antenna also if this has advantage for a particular site and does not infringe airport clearance. However, screened antenna array must be used at such installations and back-course coverage will not be possible.

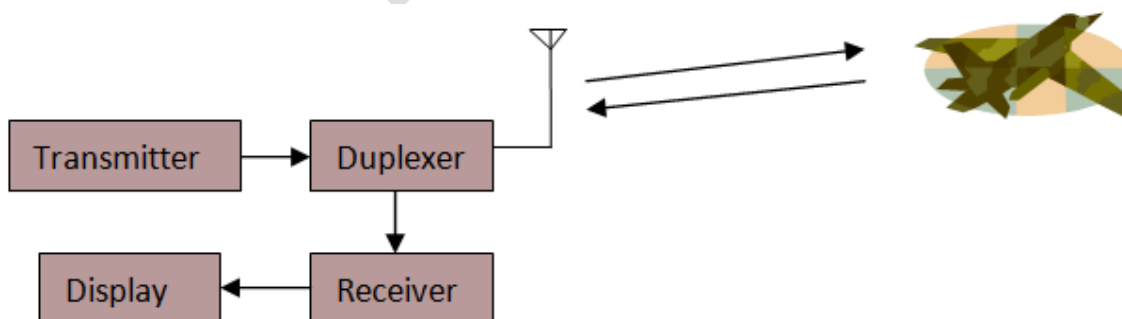
The ground surface within the site area should be as flat as possible, normally up to 1-% gradient. Beyond the site area slope should not be more than 5%. Within the airport boundary area at plus or minus 10 degrees from the antenna along the centerline of the runway, there should not be any large buildings, power lines, MF or HF antennas, or other potential reflectors.



5. AERONAUTICAL SURVEILLANCE

5.1 History of Radar

Radar is an object detection system which uses radio waves to determine the range, altitude, direction, or speed of objects. It can be used to detect aircraft, ships, spacecraft, guided missiles, motor vehicles, weather formations, and terrain. The radar dish or antenna transmits pulses of radio waves or microwaves which bounce off any object in their path. The object returns a tiny part of the wave's energy to a dish or antenna which is usually located at the same site as the transmitter. People understood the concept of radar long before scientists first built a radar system. In the late 1930's all of the countries involved in World War II rushed to create better radar systems to use in the war. Great Britain was one of the leading developers of radar in the years leading up to World War II. The research they conducted led to an early warning radar system called "Chain Home". They built radar stations around the British Isles to provide warning of an aerial invasion. This was one of the advantages which helped the outnumbered Royal Air Force defeat the German Luftwaffe during the Battle of Britain. While radar development was encouraged by wartime efforts, people were also interested in using radar as an anti-collision system. After the Titanic ran into an iceberg and sank during bad weather in 1912, people were interested in ways to avoid this happening again. Out of these all of these efforts, weather radar was developed.



A radar system has a transmitter that emits radio waves called radar signals in predetermined directions. When these come into contact with an object they are usually reflected or scattered in many directions. Radar signals are reflected especially well by materials of considerable electrical conductivity especially by most metals,

by seawater and by wet lands. Some of these make the use of radar altimeters possible. The radar signals that are reflected back towards the transmitter are the desirable ones that make radar work. If the object is moving either toward or away from the transmitter, there is a slight equivalent change in the frequency of the radio waves, caused by the Doppler Effect. Radar receivers are usually, but not always, in the same location as the transmitter. Although the reflected radar signals captured by the receiving antenna are usually very weak, they can be strengthened by electronic amplifiers. More sophisticated methods of signal processing are also used in order to recover useful radar signals.

5.3 Theory of Primary and Secondary Surveillance Radar

5.3.1. Introduction

The word RADAR was originally derived from the descriptive phrase "*Radio Detection And Ranging*". Although this phrase has for a long time been used, it seems to be an incomplete description of what Radar can be used for. The present-day RADAR can provide much more information than finding the range of an object. The fundamental principal of all Radar systems is to calculate the distance of an object from the Radar site by measuring the time a pulse of radio energy takes to travel to the object and back again. The importance of radar in aviation is that it can provide information about the precise position and velocity of the aircraft. In addition, the more complex equipment can supply other useful data, such as, velocity, identification, height, etc. Radar can contribute to the safety and surveillance of the aircraft in thick density areas. For example, near an aerodrome, where the air traffic density is very high, radar may be used to sequence the aircraft onto final approach as a final approach aid, and for separation soon after takeoff. There are two basic types of Radar system:

- Primary Radar - A system, which uses reflected radio, signals.
- Secondary Radar - A system in which radio signals transmitted from the Radar station on the ground initiates the transmission of radio signals from another station, e.g. aircraft.

➤ Use of Radar in Civil Aviation

Surveillance is one of the most important elements in aviation. Through surveillance an air traffic control post can monitor the movements of aircraft as well as can provide guidance and avert accidents. In aviation surveillance is done in two ways:

- a) Through position reporting

b) Through Radar.

Providing surveillance through position reporting by aircraft is highly unreliable and could cause misunderstandings resulting in fatal accidents. Therefore, radar has been widely used in civil aviation as one of the major surveillance tools for many years now. The prime purpose is to detect the aircraft flying within the controlled as well as uncontrolled airspace for traffic separation and control, and also for providing guidance during landing. Some of the uses of radar in aviation are as follows:

- **ASR (Airport Surveillance Radar)**
It is a medium to low power primary radar installed for surveillance and traffic separation in the airport terminal area. It works in S-band with pulse power up to 1 MW and antenna revolution around 15rpm.
- **PAR (Precision Approach Radar)**
With this radar an air traffic controller guides the approaching aircraft to take the correct approach to the airport. PAR works in the band 9000 - 9180 MHz and normally of low power.
- **ARSR (Air Route Surveillance Radar)**
As the name denotes, ARSR works as an air-route surveillance radar with relatively high coverage range of 200 to 300NM. It works in L-band (1250 - 1350 MHz) and normally placed in the air routes with high traffic density.

5.3.2 PRIMARY RADAR

5.3.2.1 General Block Diagram

The purpose of primary radar system in aviation is to present a continuous supply of useful information to the air traffic controllers on the ground regarding the range, bearing and, in some cases, elevation of the aircraft within the operational range of the radar system. Thus, every primary radar system must be capable of:

- Transmission
- Reception
- Display

A basic block diagram of the primary radar is shown in Fig.1-2. Trigger Unit, which is also called as the Master Timer, provides triggering signals in the form of a series of very brief electrical pulses at a regular interval. Each pulse fires the modulator to send a high power high voltage pulses to the transmitter. The duration of square wave pulses from the modulator is determined by certain design characteristics in the modulator. The beginning of each pulse from the modulator unit switches on the transmitter and the end of the same pulse switches it off. Thus the modulator pulses represent a kind of on/off switch for the transmitter.

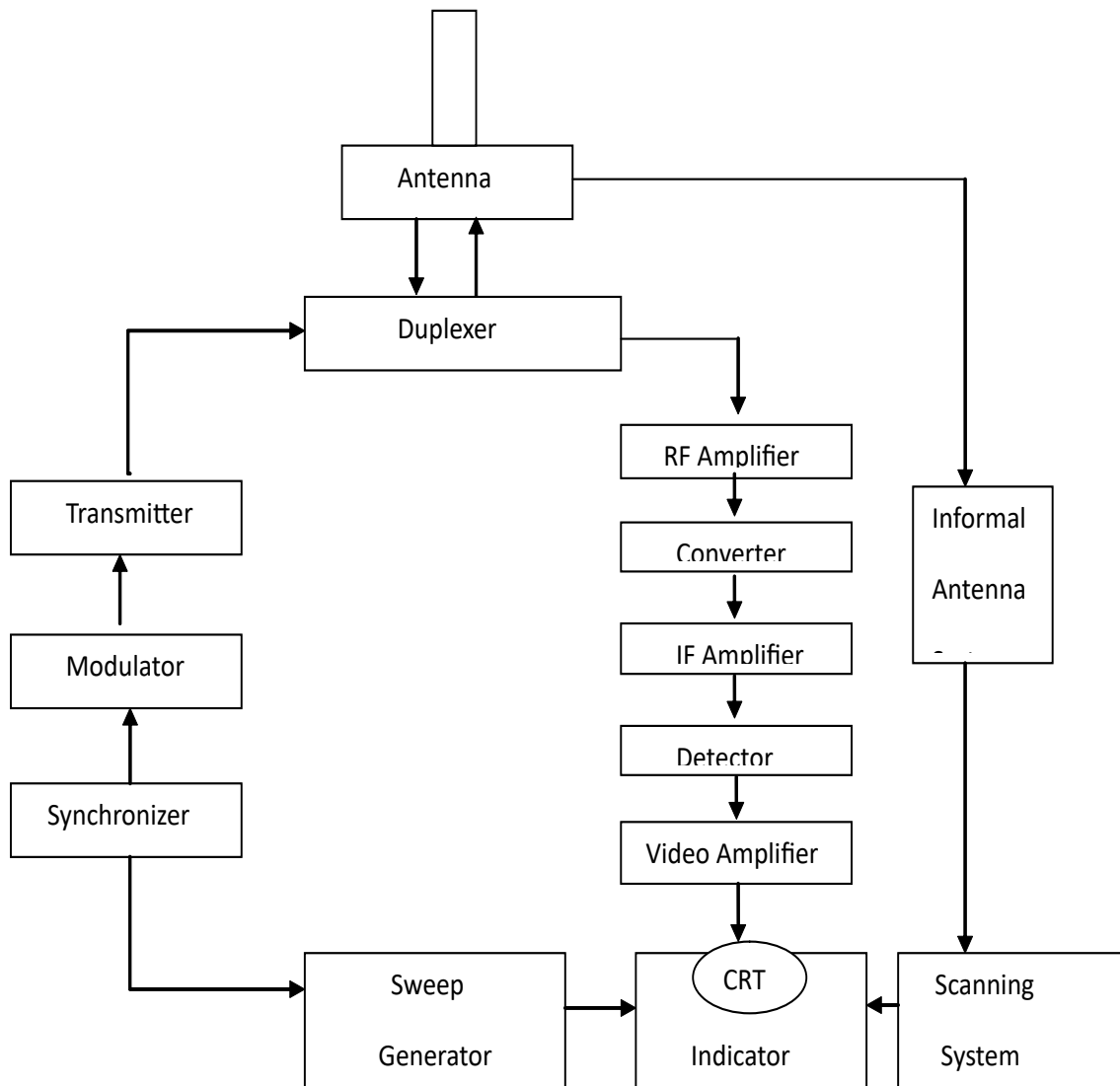


FIG. 5.2 Basic primary radar block diagram)

In primary radar, since the incoming signal has to return back from a long distance, high transmission power is desirable so as the received signal is not too weak to be detected. Therefore, the Transmitter unit consists of a Magnetron or Klystron. For low powered systems requiring smaller coverage, solid state transmitters are also used. As seen from the block diagram, the transmitter is radiating only for a very brief duration - usually in the region of millionth of a second. The output of the transmitter is a series of pulses of radar energy. Each pulse thus produced is delivered to the aerial. Electromagnetic energy from the transmitter is radiated from the aerial at a regular interval. The radiated energy is confined within a fairly narrow beam. The receiving aerial should be sensitive within the same fairly narrow angle so that echoes from the objects illuminated by the transmitter can also be detected. Therefore, normally same aerial is employed for both transmission and reception. However, because the receiver has to be very sensitive, it must be protected each time a pulse of high power is transmitted by the transmitter. This is achieved by isolating the receiver during transmission with a Transmit/Receive Switch.

If the radar system has to search an appreciable sector with a narrow beam, then the beam must be moved about in that sector. One way of achieving this movement, which is referred to as "scanning", is to rotate the whole aerial unit. Therefore, the radar antenna is being rotated at a constant speed.

The signal reception block detects energy reflected from the objects within the operational range of the radar system. The received signal, being very weak, is first of all amplified and suitably treated by the high gain low noise receiver prior to sending it to the display unit. The received signal is displayed on the screen of the display unit. As range is calculated by measuring the time between transmission and reception, it is necessary to synchronize the start of the time base with the beginning of transmitted pulse. This is done in the time base unit upon information from the trigger unit. The reference data is added in order to isolate the unwanted echoes.

5.3.2.2 Radar equation and radar range

Free space Radar equation

The ground radar systems being used in civil aviation have been designed for various specific purposes. Some of them are used for short distances whereas others for great distances. One of the important factors is to establish the maximum range. Maximum range is dependent on the characteristics of the aerial, transmitter, receiver and the size of the target. Ignoring for the time being the effect of ground on the radiation pattern from the aerial, let's examine the relationship between maximum range and the following parameters:

- R - Range of the aircraft in meters
- P_t - Transmitted power in Watts
- G - Gain of the aerial
- S - Equivalent echoing area of the aircraft
- A_r - Effective absorbing area of the aerial while receiving
- P_r - Received power in Watts

The transmitted power is the power that is transmitted for the duration of the pulse. In radar terminology this is known as the "peak power". During the period between pulses the transmitter is not generating any power. The "effective mean power" will be somewhat much smaller than the peak power and can be expressed as follows:

$$\text{Mean power} = \frac{P_t \times \text{Pulse length}}{\text{PRI}} = P_t \times \text{Pulse Length} \times \text{PRF}$$

PRI

Therefore, for example, if the peak power P_t is 1MW, Pulse length is 1 μ s and pulse repetition frequency (PRF) is 500/sec then the mean power will be:

$$\text{Mean power} = 1 \times 10^6 \times 1 \times 10^{-6} \times 500 = 500\text{W}$$

If we first of all assume that the transmitted power is radiated equally in all directions, it will be spread out progressively over great areas as it travels further away from the transmitter. As the received power density is inversely proportional to the distance from the transmitter, the power density intercepted at a distance R at the target will be:

$$\frac{P_t}{4 \pi R^2} \text{ W/m}^2$$

Now if the aerial is designed to concentrate its power in one direction only, the power density will be increased by an amount G, the gain of the antenna. Therefore the density intercepted at R will be:

$$\frac{P_t \times G}{4 \pi R^2} \text{ W/m}^2$$

Some of its energy is reflected back to the radar on the ground. The amount of reflection will depend upon the particular echoing characteristics of the aircraft and a factor equivalent to echoing area; s. Therefore, the power P_d radiated from the aircraft is given by the formula:

$$P_d = \frac{P_t \times G \times S}{4 \pi R^2} \text{ Watts}$$

As the power travels towards the radar site it is further attenuated according to the inverse square law. Therefore power density intercepted by the radar aerial P_a will be:

$$P_a = \frac{P_t \times G \times S}{16 \pi R^4} \text{ W/m}^2$$

The power actually fed into the receiver, P_r will be dependent upon the effective absorbing area, A_r of the aerial. Therefore:

$$P_r = \frac{P_t \times G \times S_x \times A_r}{16 \pi^2 R^4} \quad \text{W/m}^2$$

5.3.2.3 Radar Range

Transposing the above we could get the equation for the radar range as follows:

$$R = \sqrt[4]{\frac{P_t \times G \times S_x \times A_r}{16 \pi^2 P_r}} \quad \text{meters}$$

From the above equation it is evident that the maximum range of a primary radar is dependent on several factors, such as transmitted power, gain of the antenna, area of the reflecting object, effective absorbing area of the radar antenna and received power. However, one factor that will affect more than others is the capability of the radar receiver to receive the smallest possible signal P_r that can be identified and accepted by its threshold. If the other parameters are considered to be constant then the formula can be written as

$$R_{\max} = k \cdot \sqrt[4]{P_t}$$

From the above it is seen that if it is required to double the maximum range by increasing the transmitted power, the increase has to be 16 fold, which is not an ideal solution. Conversely, the effect of doubling the transmitted power would be to increase the maximum range by just $\sqrt[4]{2}$ times. i.e. 1.19 times, which is an increase of only 19%.

Further, the absorbing area of the antenna can be written as: $A_r = G \lambda^2 / 4\pi$

Therefore the radar equation can be now written as:

$$R_{\max} = \sqrt[4]{\frac{P_t G^2 \lambda^2 s}{64\pi^3 P_r}}$$

The above equation shows that for the fixed wavelength λ , the maximum range varies as the square root of the antenna gain, G . Therefore, if it is required to increase the range of the radar it is more practical to increase the antenna gain than to increase the transmitter power. Antenna gain can be easily increased by using special techniques available in modern-day antenna engineering.

5.3.2.4. Minimum and Maximum Range

Minimum Range

The radar energy received by the radar aerial has had to travel a distance equal to twice the range, i.e. $2R$.

Since Time $T = \frac{2R}{V}$ Where V being the velocity of light, 300,000 Km/Sec.

V

Then, $R = \frac{TV}{2}$, Hence, for pulse width of $T = 1\mu s$, $R = 150$ meters

2

The minimum range of a radar installation is determined by the period between the beginning of the transmission of the pulse and the earliest time that its return can be registered. Two factors affect this time:

1. The pulse width - It is not possible to start measuring the time until the whole pulse has been transmitted. Hence width of a pulse determines the minimum range. For example, for a pulse width of $1\mu s$ the minimum range is 150m.
2. Receiver restoration time - The time the receiver takes to recover to normal after transmission has taken place, which in turn, depends upon the design of a particular equipment.

Maximum range

In Radar the next energy pulse is only transmitted after the first one has traveled to its maximum range and had time to travel back. Therefore, the time interval between two pulses will limit the theoretical maximum range of radar.

Since $R = \frac{TV}{2}$ and $T = \frac{1}{PRF}$ Hence, $R = \frac{V}{2PRF}$ Hence $R \propto 1/PRF$

1. From the above it is seen that the maximum range attainable by a radar is inversely proportional to the pulse repetition frequency. More the pulse repetition frequency lesser the maximum attainable range by a radar installation.
2. For a particular radar system design, P_t , G and A_r will remain constant. Therefore, from the radar range equation shown hereunder it is evident that the maximum range of a radar installation will depend upon received reflected power P_r by the radar antenna. P_r is the smallest power that can be identified as a signal after passing through the receiver to the display.

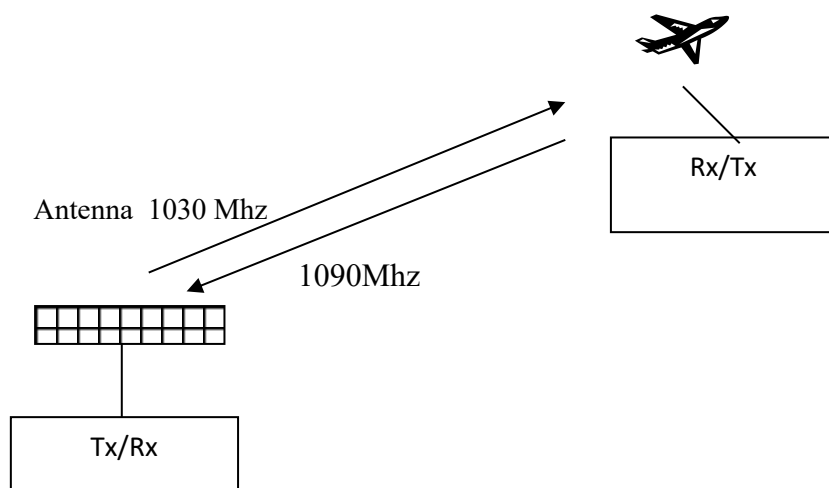
$$R = \sqrt[4]{\frac{P_t G_x S_x A_r}{16 \pi^2 P_r}} \quad \text{meters}$$

Identification of a signal maybe hampered by the presence of noise. At low frequencies the noise may arrive at the aerial mixed with the signal. However, it is the internal receiver noise that matters most. As the receiver itself generates the receiver internal noise, P_r will depend upon the quality of the receiver and will be responsible for maximum range of a particular radar for its transmitted power and aerial system. To combat the problem of receiver noise "parametric amplifiers" are commonly used in radar receivers. Research on low noise receivers continues in an attempt to achieve further improvements.

5.3.3 SECONDARY SURVEILLANCE RADAR (SSR)

5.3.3.1 General Introduction

Secondary radar is described in civil aviation as the Secondary Surveillance Radar (SSR). It consists of a ground-based transmitter/Receiver, which works in conjunction with airborne Transmitter/Receiver. The ground-based unit triggers the airborne transponder into operation when particular conditions are satisfied. SSR is used all over the world by civil aviation in only two frequencies. The ground-based transmitter, known as the "interrogator", transmits on 1030 MHz to the aircraft transceiver, known as "transponder". The transponder, having received the ground transmission on 1030 MHz, transmits back its information on 1090 MHz. The ground receiver then accepts information on 1090MHz providing air traffic controllers with vital statistics about the flight of the aircraft.



(Fig. 5-12 Basic SSR Principal)

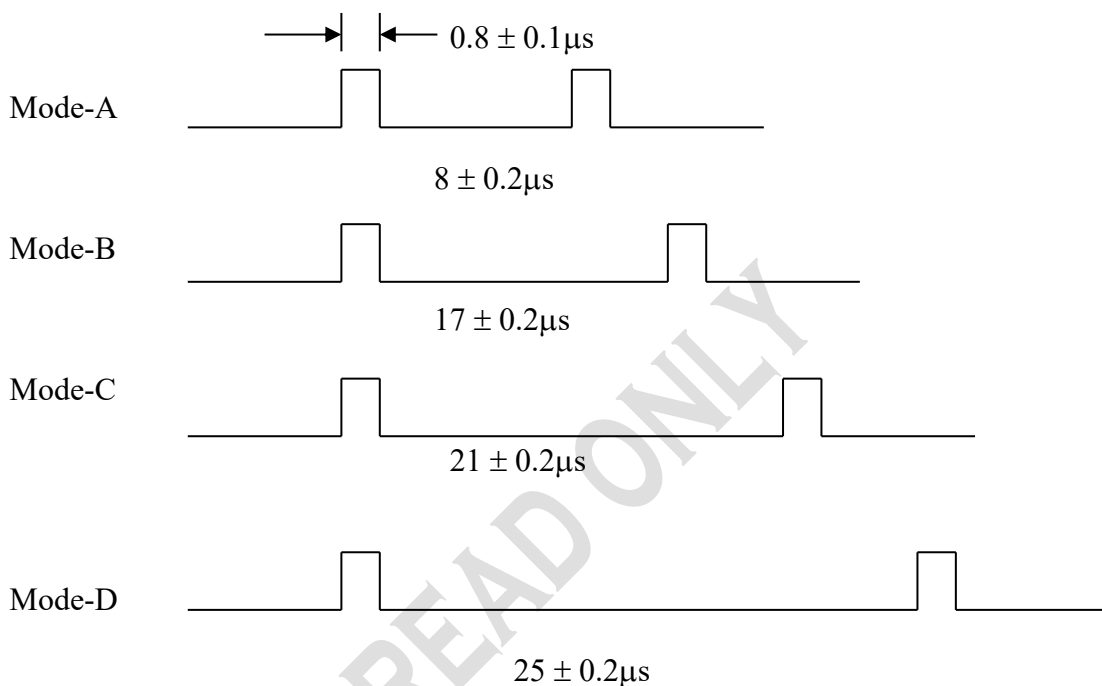
5.3.4 Comparison with Primary Radar

Primary Radar depends on the reflection of radar signal and operates on one frequency. As a consequence reflections from ground, buildings, precipitations, etc., will be fed to the display along with aircraft echoes. This can cause the display to be cluttered with unwanted signals. Also, the power output of a primary radar must be sufficiently high to allow for the two-way path and for the losses on reflection. Nevertheless, primary radar is self-contained and does not require equipment to be installed in the aircraft.

Secondary radar is independent of aircraft echoing area. In fact, because the airborne equipment has its own transmitter, much less power is needed than in primary radar for the same operational range. Also, as the airborne responses of the secondary radar is on a different frequency to the ground transmission, ground clutter and other unwanted echoes are not accepted for presentation. With suitable coding, secondary radar can provide aircraft identification without aircraft manoeuvres and thus with much less use of other communication system. The coding system may also be used to send printed information from aircraft to air-traffic controllers on ground. Unfortunately, the coding system relies on pulses being either “present” or “not present” and therefore weak signals cannot be tolerated as they may generate false information. Secondary radar also suffers from the disadvantage that it requires each aircraft to carry a transponder.

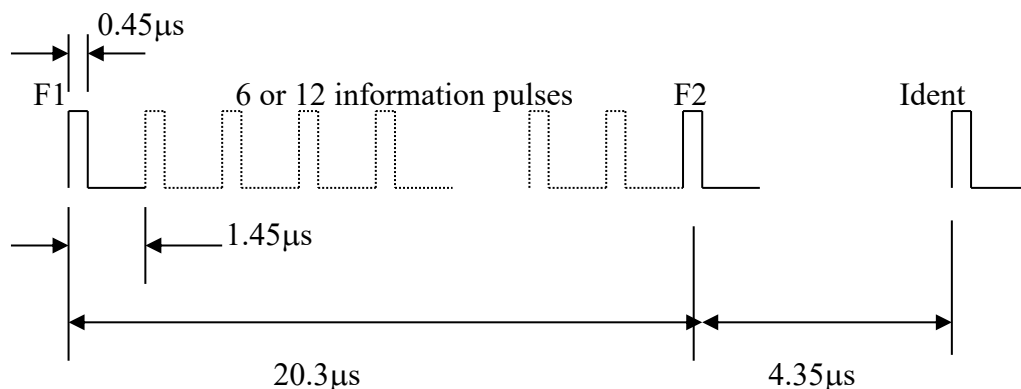
5.3.5 The Coding System

The method used in SSR for communicating information consists of the transmission and reception of pulses. The ground interrogator first transmits a pair of pulses with definite width and spacing to the aircraft. Depending upon various modes used in radar for various purposes, the pulse spacing could vary from each other. In secondary radar it is necessary to design the system to reduce the chances of transponder operation as a result of receiving spurious pulses. For this reason, two pulses with a known spacing are transmitted from the interrogator. The ground based interrogation pulses are produced differently depending upon “mode of operation” of the secondary radar.



(Fig: Various modes of SR and their pulse spacing)

After receiving the interrogation pair of pulses from the ground interrogator, the aircraft transponder transmits back a reply to ground. In the process of replying, the aircraft transponder does not reply to any other signals at least for the duration of the reply pulses train. It is known as receiver dead time of radar transponder. This dead time does not last more than 125 microseconds after transmission of the last reply pulses. The aircraft transponder recognizes the interrogation mode by the time spacing between two pulses. The transponder reply, as illustrated below, consists of a train of pulses each of which is $0.45 \mu\text{s}$ wide and $1 \mu\text{s}$ apart.



(Fig. 5.14 Reply Pulse Train from transponder)

The train of pulses is built up as follows:

Two framing pulses $20.3\mu\text{s}$ apart (F1 and F2). The first frame pulse is always the first pulse in the train. The information pulses lie between two frame pulses and composed of either 6 or 12 pulses. The information signal is formed in binary codes, which means either presence or absence of pulses. If six pulse positions are used then the number of codes available is $2^6 = 64$. For 12 positions the number of codes available is $2^{12} = 4096$. The identity pulse is transmitted only when the switch is activated in the aircraft at the request of the air traffic controller. The identity pulses are then transmitted automatically for a period of twenty seconds.

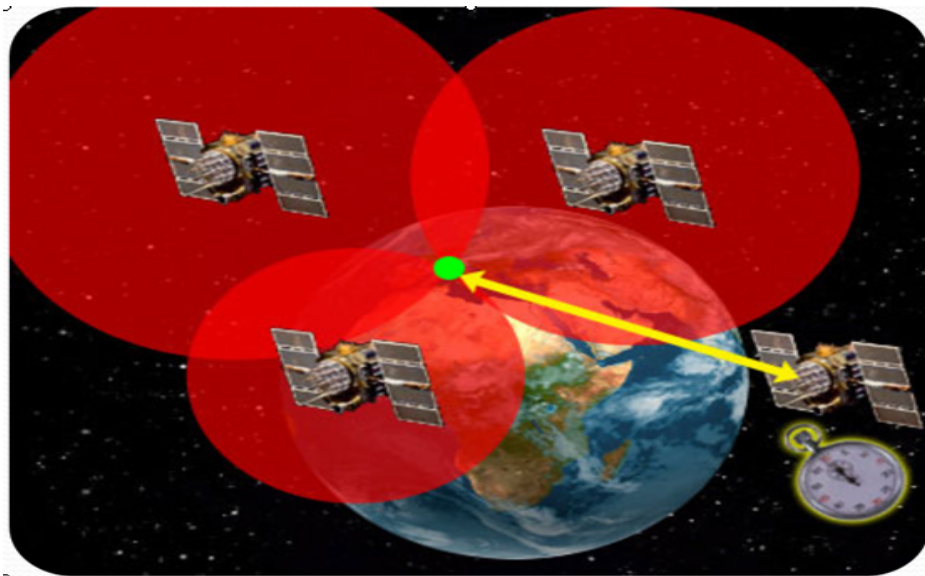
6.3. Global Positioning System

6.3.1. Introduction

The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites. GPS was created and realized by the U.S Department of Defense (DoD) and was originally run with 24 satellites. Roger L. Easton is generally credited as its inventor. It consists of satellites, control and monitor stations, and receivers. GPS receivers take information transmitted from the satellites and uses triangulation to calculate a user's exact location.

How the Global Positioning System Works

The basis of the GPS is a constellation of satellites that are continuously orbiting the earth. These satellites, which are equipped with atomic clocks, transmit radio signals that contain their exact location, time, and other information. The radio signals from the satellites, which are monitored and corrected by control stations, are picked up by the GPS receiver. A GPS receiver needs only three satellites to plot a rough, 2D position, which will not be very accurate. Ideally, four or more satellites are needed to plot a 3D position, which is much more accurate. The receiver computes the position using trilateration. Trilateration is the process of determining absolute or relative locations of points by measurement of distances, using the geometry of circles, spheres or triangles.



GPS Functionality

It consists of three segments:

Space Segment — Satellites orbiting the earth

The space segment consists of 29 satellites circling the earth every 12 hours at 12,000 miles in altitude. This high altitude allows the signals to cover a greater area. The satellites are arranged in their orbits so a GPS receiver on earth can receive a signal from at least four satellites at any given time. Each satellite contains several atomic clocks. The satellites transmit low radio signals with a unique code on different frequencies, allowing the GPS receiver to identify the signals. The main purpose of these coded

signals is to allow the GPS receiver to calculate travel time of the radio signal from the satellite to the receiver. The travel time multiplied by the speed of light equals the distance from the satellite to the GPS receiver.

Control Segment — The control and monitoring stations

The control segment tracks the satellites and then provides them with corrected orbital and time information. The control segment consists of five unmanned monitor stations and one Master Control Station. The five unmanned stations monitor GPS satellite signals and then send that information to the Master Control Station where anomalies are corrected and sent back to the GPS satellites through ground antennas.

User Segment — The GPS receivers owned by civilians and military

The user segment consists of the users and their GPS receivers. The number of simultaneous users is limitless.

How GPS Determines a Position

The GPS receiver uses the following information to determine a position:

Precise location of satellites

When a GPS receiver is first turned on, it downloads orbit information from all the satellites called an almanac. This process, the first time, can take as long as 12 minutes; but once this information is downloaded, it is stored in the receiver's memory for future use.

Distance from each satellite

The GPS receiver calculates the distance from each satellite to the receiver by using the distance formula: $\text{distance} = \text{velocity} \times \text{time}$. The receiver already knows the velocity, which is the speed of a radio wave or 186,000 miles per second (the speed of light). To determine the time part of the formula, the receiver times how long it takes for a signal from the satellite to arrive at the receiver. The GPS receiver multiplies the velocity of the transmitted signal by the time it takes the signal to reach the receiver to determine distance.

Triangulation to determine position

The receiver determines position by using triangulation. When it receives signals from at least three satellites the receiver should be able to calculate its approximate position (a 2D position). The receiver needs at least four or more satellites to calculate a more

accurate 3D position. The position can be reported in latitude/longitude, UTM, or other coordinate system.

Sources of Errors

The GPS is not a perfect system. There are several different types of errors that can occur when using a GPS receiver, for example:

User mistakes

User mistakes account for most GPS errors; and a GPS receiver has no way to identify and correct these mistakes. Common examples of user mistakes include:

- Inputting incorrect information into a GPS receiver, such as the datum, and when creating a waypoint.
- Unknowingly relying on a 2D position instead of a 3D position for determining position coordinates. This mistake can result in distance errors in excess of a mile. The signal from the satellite may be blocked by buildings, terrain, electronic interference, and sometimes dense foliage. A GPS receiver needs a fairly clear view of the sky to operate.
- The human body can cause signal interference. Holding a GPS receiver close to the body can block some satellite signals and hinder accurate positioning. If a GPS receiver must be hand held without benefit of an external antenna, facing to the south can help to alleviate signal blockage caused by the body because the majority of GPS satellites are oriented more in the earth's southern hemisphere.

Multipath interference

Multipath interference is caused by the satellite signal reflecting off of vehicles, buildings, power lines, water and other interfering objects. Multipath is difficult to detect and sometimes impossible for the user to avoid or for the receiver to correct. When using a GPS receiver in a vehicle place the external antenna on the roof of the vehicle to eliminate most signal interference caused by the vehicle. If the GPS receiver is placed on the dashboard there will always be some multipath interference.

Satellite and receiver clock errors

These can be slight discrepancies in the satellite's atomic clocks which may cause slight position errors in the GPS receiver. Errors are monitored and corrected by the Master Control Station.

Orbit errors

Satellite orbit pertains to the altitude, position, and speed of the satellite. Satellite orbits vary due to gravitational pull and solar pressure fluctuations. Orbit errors are also monitored and corrected by the Master Control Station.

Satellite geometry

The location of GPS satellites in relation to a GPS receiver on the ground can impact the receiver's ability to triangulate a 3D position. The quality of a receiver's triangulated position improves the further apart GPS satellites are located from each other in the sky above the receiver. The quality decreases if the satellites are grouped close together in the sky above the receiver.

Atmospheric interference

The atmosphere can slow or speed up the satellite signal. Fortunately, error caused by atmospheric conditions (ionized air, humidity, temperature, pressure) has been reduced with the implementation of the Wide Area Augmentation System (WAAS).

Application

1. Military

- Military GPS user equipment has been integrated into fighters, bombers, tankers, helicopters, ships, submarines, tanks, jeeps, and soldiers' equipment.
- In addition to basic navigation activities, military applications of GPS include target designation of cruise missiles and precision-guided weapons and close air support.

2. Civilian

Automobiles are often equipped GPS receivers.

- They show moving maps and information about your position on the map, speed you are traveling, buildings, highways, exits etc.

For aircraft, GPS provides

- Continuous, reliable, and accurate positioning information for all phases of flight on a global basis, freely available to all.

3. Agriculture

- GPS provides precision soil sampling, data collection, and data analysis, enable localized variation of chemical applications and planting density to suit specific areas of the field.

6.4. Global Orbiting Navigation Satellite System (GLONASS)

The Global Orbiting Navigation Satellite System (GLONASS) is based on a constellation of active satellites which continuously transmit coded signals in two frequency bands, which can be received by users anywhere on the Earth's surface to identify their position and velocity in real time based on ranging measurements. The system is a counterpart to the United States Global Positioning System (GPS) and both systems share the same principles in the data transmission and positioning methods. GLONASS is managed for the Russian Federation Government by the Russian Space Forces and the system is operated by the Coordination Scientific Information Center (KNITs) of the Ministry of Defense of the Russian Federation.

The operational space segment of GLONASS consists of 21 satellites in 3 orbital planes, with 3 on-orbit spares. The three orbital planes are separated 120 degrees, and the satellites within the same orbit plane by 45 degrees. Each satellite operates in circular 19,100 km orbits at an inclination angle of 64.8 degrees and each satellite completes an orbit in approximately 11 hours 15 minutes.

The ground control segment of GLONASS is entirely located within former Soviet Union territory. The Ground Control Center and Time Standards is located in Moscow and the telemetry and tracking stations are in St. Petersburg, Ternopol, Eniseisk, Komsomolsk-na-Amure.

The first GLONASS satellites were launched into orbit in 1982. Two Etalon geodetic satellites were also flown in the 19,100 km GLONASS orbit to fully characterize the gravitational field at the planned altitude and inclination. The original plans called for a complete operational system by 1991, but the deployment of the full constellation of satellites was not completed until late 1995 / early 1996. GLONASS was officially declared operational on September 24, 1993 by a decree of the President of the Russian Federation.

Reading